Behavioral/Cognitive

Neurochemical Predictors of Generalized Learning Induced by Brain Stimulation and Training

Shane E. Ehrhardt,[1] Yohan Wards,[1] Reuben Rideaux,[2,3] Małgorzata Marjańska,[4] Jin Jin,[5,6] Martijn A. Cloos,[5] Dinesh K. Deelchand,[4] Helge J. Zöllner,[7] Muhammad G. Saleh,[8] Steve C. N. Hui,[7] Tonima Ali,[9,10] Thomas B. Shaw,[5,11] Markus Barth,[5,11] Jason B. Mattingley,[1,2,12] Hannah L. Filmer,[1] and Paul E. Dux[1]

[1]School of Psychology, The University of Queensland, St Lucia, Queensland 4072, Australia, [2]Queensland Brain Institute, The University of Queensland, St Lucia, Queensland 4072, Australia, [3]School of Psychology, The University of Sydney, Sydney, New South Wales 2050, Australia, [4]Department of Radiology, Centre for Magnetic Resonance Research, University of Minnesota, Minneapolis, Minnesota 55455, [5]Centre for Advanced Imaging, The University of Queensland, St Lucia, Queensland 4072, Australia, [6]Siemens Healthcare Pty Ltd., Brisbane, Queensland 4006, Australia, [7]The Russell H. Morgan Department of Radiology and Radiological Science, The Johns Hopkins University School of Medicine, Baltimore, Maryland 21287, [8]Department of Diagnostic Radiology and Nuclear Medicine, University of Maryland School of Medicine, Baltimore, Maryland 21201, [9]School of Biomedical Engineering, The University of Sydney, Sydney, New South Wales 2050, Australia, [10]Brain and Mind Centre, The University of Sydney, Sydney, New South Wales 2050, Australia, [11]School of Information Technology and Electrical Engineering, The University of Queensland, St Lucia, Queensland 4072, Australia, and [12]Canadian Institute for Advanced Research (CIFAR), Toronto, Ontario M5G 1M1, Canada

Methods of cognitive enhancement for humans are most impactful when they generalize across tasks. However, the extent to which such "transfer" is possible via interventions is widely debated. In addition, the contribution of excitatory and inhibitory processes to such transfer is unknown. Here, in a large-scale neuroimaging individual differences study with humans (both sexes), we paired multitasking training and noninvasive brain stimulation (transcranial direct current stimulation, tDCS) over multiple days and assessed performance across a range of paradigms. In addition, we varied tDCS dosage (1.0 and 2.0 mA), electrode montage (left or right prefrontal regions), and training task (multitasking vs a control task) and assessed GABA and glutamate concentrations via ultrahigh field 7T magnetic resonance spectroscopy. Generalized benefits were observed in spatial attention, indexed by visual search performance, when multitasking training was combined with 1.0 mA stimulation targeting either the left or right prefrontal cortex (PFC). This transfer effect persisted for ∼30 d post intervention. Critically, the transferred benefits associated with right prefrontal tDCS were predicted by pretraining concentrations of glutamate in the PFC. Thus, the effects of this combined stimulation and training protocol appear to be linked predominantly to excitatory brain processes.

Key words: **cognitive training; glutamate; tDCS; transfer; ultrahigh field MRS; visual search**

Significance Statement

Despite the general public's fascination with cognitive training, performance benefits rarely extend beyond the trained task, that is, "transfer." Our study examines the impact of combining executive function training and transcranial direct current stimulation (tDCS) on human cognitive performance and identifies a functional neural metabolite marker (glutamate concentrations in the prefrontal cortex assessed via 7T MR spectroscopy) that predicts outcomes. In the largest study of its kind to date (178 individuals), we find generalized performance benefits induced by frontal tDCS for an untrained spatial attention task. Further, the degree of transfer correlated with concentrations of glutamate in the frontal cortex. Thus, excitatory neural processes in this region are implicated in the transfer of paired stimulation and training benefits.

Received Sept. 5, 2023; revised Jan. 22, 2024; accepted March 8, 2024.

Author contributions: S.E.E., Y.W., M.M., J.J., M.A.C., D.K.D., S.C.N.H., T.A., M.B., J.B.M., H.L.F., and P.E.D. designed research; S.E.E., Y.W., H.L.F., and P.E.D. performed research; T.B.S. contributed unpublished reagents/analytic tools; S.E.E., Y.W., R.R., M.M., J.J., D.K.D., H.J.Z., M.G.S., and H.L.F. analyzed data; S.E.E., J.B.M., H.L.F., and P.E.D. wrote the paper.

We thank research radiographers, Nicole Atcheson and Aiman Al-Najjar, and research assistants Kali Chidley and Zoie Nott. This work was supported by the Australian Research Council Discovery Project Grant DP180101885 (P.E.D., J.B.M.), Australian Research Council Discovery Project Grant DP210101977 (P.E.D., H.L.F., J.B.M.), Australian Research Council Discovery Early Career Research Award DE210100790 (R.R.), Australian Research Council Discovery Early Career Research Award DE190100299 (H.L.F.), Australian Research Council Future Fellowship Grant FT200100329 (M.A.C.), Department of Defence (Human Performance Research (HPR) Network Partnership (P.E.D., H.L.F., J.B.M.), Australian Government Research Training Program Scholarship (S.E.E., Y.W.), National Institutes of Health Grants BTRC P41 EB027061 and P30 NS076408 (D.K.D., M.M.), National Health and Medical Research Council (Australia) Investigator Grant GNT2010141 (J.B.M.), and Motor Neurone Disease Research Australia (MNDRA) Postdoctoral Research Fellowship PDF2112 (T.B.S.).

The authors declare no competing financial interests.

Correspondence should be addressed to Shane E. Ehrhardt at s.ehrhardt@uq.edu.au.

https://doi.org/10.1523/JNEUROSCI.1676-23.2024

Copyright © 2024 the authors

Introduction

Learning is most impactful when it generalizes across tasks/processes as such "transfer" maximizes intervention efficiency. However, the efficacy of cognitive training paradigms in generating such transfer is debated. While highly impactful work suggests that effortful training can generalize gains across psychological operations (Jaeggi et al., 2008, 2011; Klingberg, 2010; Anguera et al., 2013), there have been a number of prominent failures to find such effects (Owen et al., 2010; Redick et al., 2013). Thus, the evidence that training-based interventions alone can improve performance beyond the trained-on task (i.e., far transfer) is mixed (Shipstead et al., 2012; Melby-Lervåg et al., 2016; Simons et al., 2016; Sala and Gobet, 2017, 2019; Bigand and Tillmann, 2022).

Recently, promising transfer results have been observed when pairing cognitive training with noninvasive brain stimulation (Looi et al., 2016; Trumbo et al., 2016; Filmer et al., 2017a,b; Ehrhardt et al., 2022). The potential for this approach to have an applied impact is considerable, as the most common brain stimulation method—transcranial direct current stimulation (tDCS)—is low-cost, easy to implement, and without significant side effects. tDCS involves running a small current typically between two electrodes (a positive anode and a negative cathode), where one or both can be placed on the scalp, targeting specific underlying brain regions (Priori et al., 1998; Nitsche and Paulus, 2000, 2001), to establish the causal neural substrates of behavior (Filmer et al., 2013; Miniussi et al., 2013; Wörsching et al., 2016; Reinhart et al., 2017). While the neurophysiological consequences of stimulation are yet to be fully elucidated (Filmer et al., 2014, 2020b; Stagg, 2014; Jackson et al., 2016; Bikson et al., 2019), there is evidence that tDCS affects neurotransmitter levels and its effects vary with baseline neurotransmitter concentrations (Rango et al., 2008; Stagg et al., 2009; Bachtiar et al., 2018; Patel et al., 2019; Filmer et al., 2019a; Jia et al., 2022). Specifically, the key neurochemicals γ-aminobutyric acid (GABA, inhibitory) and glutamate (excitatory)—and the balance between the two, the so-called excitation/inhibition, or *E*/*I* balance or ratio—appear to be important for stimulation efficacy for single-session paradigms (Filmer et al., 2019a).

Of import for the present study, pairing anodal tDCS over the left prefrontal cortex (PFC) with single-task decision-making (Filmer et al., 2017b), or multitasking training (Filmer et al., 2017a), enhanced performance on an unrelated visual search task (Filmer et al., 2017b), even when the trained-upon task itself was not influenced by stimulation (Filmer et al., 2017a). Given these findings, here we focus our assessment of transfer on performance on a visual search paradigm. However, we also include a range of cognitive measures to assess the extent to which training and stimulation may influence performance. In addition, a challenge for studies employing noninvasive brain stimulation approaches, including tDCS, is that their efficacy often varies across individuals (Krause and Cohen Kadosh, 2014; López-Alonso et al., 2014; Katz et al., 2017a; Laakso et al., 2019; Filmer et al., 2019a). Understanding the neural basis of this variability is crucial if the full benefits of tDCS are to be realized.

To wit, the factors affecting tDCS efficacy in multisession training and transfer remain unclear. Here, we examined if baseline levels of GABA and glutamate are associated with performance enhancement via combined tDCS and cognitive training. In a preregistered study, we applied rigorous controls across multiple training/stimulation groups and used ultrahigh field 7 Tesla (T) MRS to quantify neurochemical concentrations in targeted brain areas (Figs. 1, 2). We hypothesized that multitasking training combined with left hemisphere tDCS would yield significant improvements in spatial attention, as assessed via visual search, and that these improvements would be predicted by underlying neurochemical concentrations. To preview the results, pairing multitasking training with stimulation of the left or right PFC led to reliable transfer to visual search, and this effect was associated with baseline glutamate concentrations. The transfer effect, specific to a 1.0 mA dosage, persisted ∼30 d post-training.

Materials and Methods

Overview

We trained 178 healthy individuals (18–40 years) on a multitasking paradigm previously found to generate transfer (Filmer et al., 2017a). Participants were split across five conditions to test the role of stimulation region and intensity, and the specificity of the task paired with stimulation, on outcomes. Participants attended 10 sessions—three neuroimaging sessions, a pre-, post-, and follow-up behavioral assessment session, and four training sessions on consecutive days (Fig. 1).

Experimental design

The study design and analyses (unless otherwise stated) were preregistered at the open science framework (OSF; https://tinyurl.com/5h8u72j5). The experiment consisted of ten sessions: pre-, post-, and follow-up (relative to the intervention sessions) test sessions, four combined tDCS-plus-training intervention sessions, and three imaging sessions—baseline MRS and structural scans as well as pre- and post-training functional and structural scans. Multiple other imaging modalities were acquired but are not presented here as they are beyond the scope of the present work: specifically, task-based fMRI, resting-state fMRI, diffusion-weighted imaging, fluid-attenuated inversion recovery, and quantitative susceptibility mapping (see Fig. 1 for experimental timeline). The pre-, post-, and follow-up test sessions consisted of a comprehensive battery of nine cognitive tasks aimed at tapping distinct cognitive processes. Each will be described in detail in the subsequent section. All training, pre-, post-, and follow-up sessions were run on Mac minicomputers (2014 model; 2.8 GHz Intel Core i5, OSX High Sierra v 10.13.6 and displayed on 24 in ASUS VG248 monitors, 144 Hz refresh rate). All of the tasks were run with MATLAB 2016b (MathWorks) using custom code and Psychtoolbox (Brainard, 1997; Pelli, 1997; see http://psychtoolbox.org). Participants were set up to have their heads approximately 57 cm away from the screen. Care was taken to ensure that they remained in approximately this position while completing tasks; however, they were not interrupted during a task to avoid compromising performance, and a chin rest was not used. We used a between-subject design with participants split equally across five groups (note: all stimulation "targets" are in reference to the anode electrode). Two groups, the sham left PFC and 1.0 mA left PFC stimulation paired with multitasking training were double-blinded throughout the entirety of the study. The other three groups serve as active controls: (1) a dosage group, 2.0 mA applied to the left PFC combined with single-/dual-task training; (2) a different montage group, 1.0 mA applied to the right PFC combined with single-/dual-task training; (3) and a different training task group, namely, 1.0 mA applied to the left PFC combined with rapid serial visual presentation (RSVP) task training (Fig. 1). The order of task completion was pseudorandomized and counterbalanced across all participants and groups to minimize any potential influence of task order on baseline performance measures.

Participants

Despite employing a Bayesian analysis approach, we derived our sample size and stopping rule via a power analysis with G*Power using conservative effect sizes of the behavioral effects of tDCS on multitasking training (approximately Cohen's *D* of 0.3), which was derived from our lab's previous studies (Garner and Dux, 2015; Filmer et al., 2017a; Ehrhardt et al., 2021). We calculated that we would require 33 participants per group for the between-group comparisons to achieve 95% power. For the individual difference's component, we calculated that we would require 46 participants in each group to achieve 95% power. Therefore, we aimed for a maximum sample size of 250 participants (50 per group: allowing

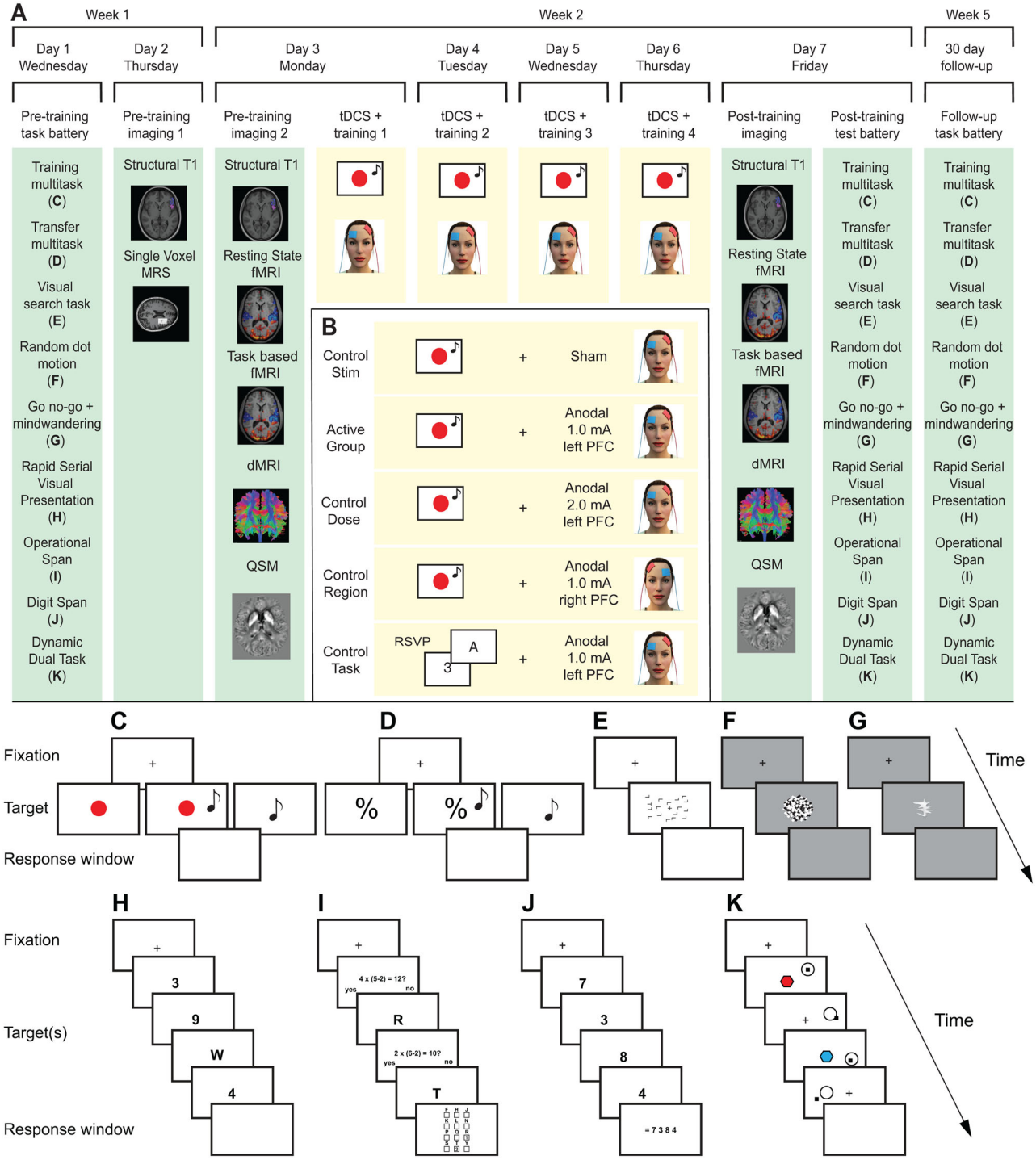

Figure 1. Experimental design. ***A***, Participants completed 10 sessions on eight separate training days spread over 5 weeks. Three sessions were dedicated to the cognitive battery used to assess training and transfer across multiple cognitive domains before, directly after, and ∼30 d post-training. Three MRI sessions were completed with only the MRS findings as the focus of the research here. Participants also completed 4 d of combined tDCS and training, with four groups training on a multitasking protocol and one active control group training on a RSVP task. ***B***, Experimental groups. ***C–K***, Task schematics for each of the paradigms included in the battery. ***C***, The multitasking protocol, also used as the main training task, which consisted of three different visual stimuli and three auditory tones presented either alone (single task) or together (dual task). ***D***, Transfer multitasking protocol, identical to the training task but with different visual and auditory stimuli. ***E***, Visual search task. ***F***, Random dot motion kinematogram task. ***G***, Go/no-go task. ***H***, RSVP task, also used as the control training group task. ***I***, Operation span task. ***J***, Forward and reverse digit span task. ***K***, Dynamic dual task, continuous visuomotor tracking task combined with a perceptual discrimination task.

for meaningful between- and within-subject analyses). Due to unforeseen circumstances (the COVID-19 pandemic and associated lockdowns), we reached a final sample of 207 participants. We had implemented a stopping date regardless of the number of participants recruited; 28th of September 2021; however, data were collected by 15th January 2022 to meet our minimum target sample size for the between-subject comparisons. This decision to continue testing was made without examining the data. Twenty-nine subjects were excluded for reasons outlined below, leaving a final sample of 178 participants (119 females) aged 18–40 years ($M = 22.86$, $SD = 3.93$ years), who

Table 1. Demographics by group

	Sham	1.0 mA left PFC MT	1.0 mA right PFC MT	1.0 mA left PFC RSVP	2.0 mA left PFC MT	Total
N	36	35	35	36	36	178
Age	22.56	23.09	23.26	23.22	22.19	22.86
Sex (M/F)	11/25	11/24	12/23	12/24	13/23	59/119

Values represent mean age and count for *N* (sample) and sex.

completed sessions 1–9 (see the experimental design in Fig. 1). Of those, 167 completed session 10 as well (30 d follow-up). See Table 1 for demographic breakdown by group.

Exclusions and group randomization

Subjects were excluded from the final sample for a variety of reasons. Six subjects were informed of incidental findings (all pineal gland cysts) after their initial MRI, prior to any brain stimulation sessions, and were excluded for precautionary reasons. The remaining reasons for exclusions were as follows: ten participants for not tolerating the MRI scans (nausea/dizziness, claustrophobia, anxiety, noise, session too long); three participants for being unable to complete sessions due to COVID-related lockdowns; three participants failed to reach 60% accuracy in the multitasking paradigms; two participants failed to attend all sessions; one participant had to leave the country for personal reasons; one participant was removed for inappropriate behavior; one participant withdrew after the first training session as they found the tDCS uncomfortable (though no adverse effects were noted); one participant took a psychoactive medication prior to a tDCS session; and one participant for drastically reduced accuracy in a training session. For the 30 d follow-up session, eight participants couldn't make it due to COVID-19 lockdowns, and three participants failed to return for testing.

Prior to any testing sessions, all participants passed a safety screening questionnaire, an MRI safety questionnaire, and a small quiz to ensure they read and understood the information sheet that outlined the techniques being used in the study (i.e., tDCS and MRI) in advance of participation. Participants were then excluded prior to any testing based on criteria including any history of brain trauma (e.g., concussion), current use of psychoactive medications, and individual or family history of epilepsy. The study was approved by The University of Queensland Human Research Ethics Committee. All participants provided written informed consent each session and were paid for their time at a rate of $20 AUD/h (roughly ∼$380 if they completed all 10 sessions). Regardless of whether participants finished the study or not, they were reimbursed for their time.

After completion of session one, participants were allocated into their groups using a custom automated group assignment algorithm to limit the likelihood of baseline group differences. The algorithm pseudorandomized the allocation of participants to groups (as in Ehrhardt et al., 2021) based on the key characteristics of age, sex, time of day (am or pm) of intervention, single- and dual-task reaction times, and the order in which participants completed their tasks.

Brain stimulation parameters

Stimulation was applied via a NeuroConn stimulator using two 5×5 cm^2 saline-soaked sponges containing rubber electrodes, held in place with rubber straps. For groups with left PFC as the target region, the target (anodal) electrode was placed 1 cm posterior to F3 (Filmer et al., 2017a,b; Ehrhardt et al., 2021) based on the 10–20 EEG system (Jasper, 1958), with the cathode placed over the right supraorbitofrontal cortex. Many previous studies investigating executive function used this configuration (Dedoncker et al., 2016). The group who received right PFC stimulation had the anode placed 1 cm posterior to F4 with the cathode placed over the left supraorbitofrontal cortex (mirroring the left hemisphere montage, Fig. 1). Participants received 13 min of online stimulation (Filmer et al., 2017a) during each training session, with a 30 s linear ramp up and down. Given previous evidence that tDCS outcomes have been influenced by variability arising from factors such as time of day (Ridding and Ziemann, 2010), all participants completed each of their sessions at roughly the same time of day (±2 h). Most tDCS studies (95%) use 1.0 and 2.0 mA (Bikson et al., 2016); as such, we chose these two intensities to help generalize the dosage and montage to other findings. We have also observed differential effects of stimulation intensity on multitasking performance (Ehrhardt et al., 2021) and decision processes (Ehrhardt et al., 2022)

Multitask training

The main training task was a sensory–motor decision-making task (Filmer et al., 2017a; Ehrhardt et al., 2021). The stimuli consisted of either a centrally presented colored circle (either a red, RGB 237 32 36; dark green, RGB 10 130 65' or dark blue, RGB 44 71 151, subtending approximately 2.7° of visual angle) and/or one of three complex tones (as used in Dux et al., 2006). Participants responded by pressing key(s) on a keyboard, with each stimulus mapped to a specific key [A, S, D (left hand) J, K, L (right hand); with the index fingers on the D and J keys, respectively]. There were three different trial types: single-visual (a colored circle; red, green, or blue), single-auditory (one of three complex tones), or a combined visual–auditory dual-task trial (one colored circle and one complex tone, presented simultaneously). For dual-task trials, both stimuli were presented simultaneously (0 ms SOA) in line with previous work studying multitasking (Dux et al., 2009). Trials proceeded as follows: a fixation square (0.4° diameter) was centrally presented for either 600 or 1,000 ms, followed by a 200 ms stimulus presentation. For visual stimuli, these were presented centrally; auditory stimuli were presented in stereo through Sure SRH440 over-ear headphones. Participants had 2,200 ms from stimulus onset to respond. Response mappings were counterbalanced across participants and groups, where half of the participants used their left hands for the auditory task and right hands for the visual task, and the other half vice versa. Participants were required to respond as quickly and as accurately as possible. At the pretraining

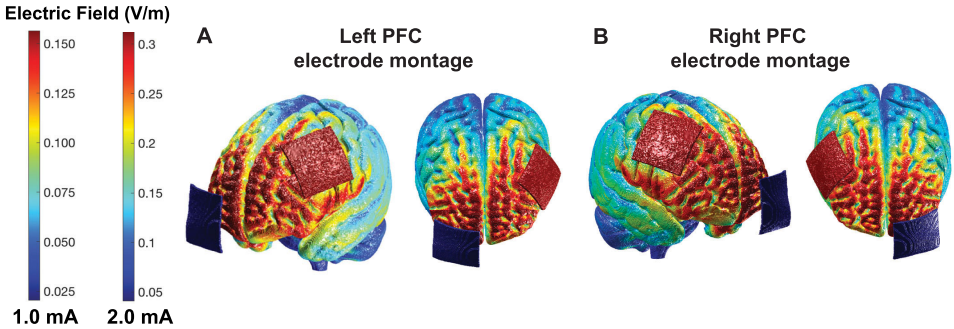

Figure 2. Electric field modeling. ***A***, Current modeling for the four left PFC stimulation groups. The anode targets 1 cm posterior to F3 (targeting left posterior lateral PFC), with reference electrode targeting Fp2 (right supraorbitofrontal cortex) on a standardized head. ***B***, Current modeling for the right PFC stimulation group. The anode targets 1 cm posterior to F4 (targeting right posterior lateral PFC), with the reference electrode targeting Fp1 (left supraorbitofrontal cortex) on a standardized head; ROAST NY head (Huang et al., 2016, 2019). The distribution of the e-field is assumed to be the same with both stimulation intensities (1 or 2 mA).

session, participants practiced both single and dual tasks until a 70% accuracy cutoff for the dual-task condition was reached to minimize exclusion rates due to poor performance. The initial practice consisted of three blocks: the first two blocks contained 15 trials of each single task (auditory and visual), respectively, while the third block had 30 trials with all trial types (single-visual, single-auditory, or dual-task) with each trial type being presented 10 times, in random order. If 70% accuracy wasn't reached, participants would repeat the third block until 70% dual-task accuracy was reached. The number of practice blocks completed by the multitasking training groups was equivalent across pretraining, post-training, and follow-up sessions and training sessions 1–4 [one-way ANOVA: Bayes factor (BF_{10}) = 0.008, $p = 0.539$]. There were no differences in pre-, post-, and follow-up training sessions when including the control training task group (RSVP training: who trained on an RSVP task for training sessions 1–4; one-way ANOVA: $BF_{10} = 0.017$, $p = 0.545$). In all sessions (pre-, post-, follow-up, and training sessions), participants completed 240 trials, split equally between the three trial types (single-visual, 80; single-auditory, 80; dual-task, 80) with a 30 s break at the halfway point. Prior to each subsequent session, participants completed Block 3 of the practice session mentioned above to refresh the response mappings (10 trials of each: single-visual, single-auditory, and dual-task). Training on this task aimed to increase the speed (i.e., reduce information processing time) at which participants can execute a single or multiple decision(s). Each 240-trial session lasted ∼13 min. This task was also in the pre-, post-, and follow-up testing sessions for the training control task group that trained on a RSVP task described below.

Control training task

To assess if any potential benefits of combining stimulation and training are related specifically to multitasking training, one active control group trained on a selective attention training control task: RSVP (see Dux and Marois (2009) for a review). Participants were presented with a fixation square of 0.4° diameter that lasted for a randomly determined duration of either 0.2, 0.3, 0.4, 0.5, or 0.6 s. This was followed by a rapid stream of seven distractor numbers with participants having to identify a single target "letter" embedded within the stream (i.e., 4, 2, W, 5, 3, 8, 9, 7). The height of the numerical and alphabetical stimuli was approximately 0.8° of visual angle. Responses were made via keypress on a standard keyboard corresponding to the presented letter (possible letters presented: "A," "B," "C," "D," "E," "F," "G," "H," "J," "K," "M," "N," "P," "R," "S," "T," "W," "Y," "Z"). Letter position was randomly allocated throughout the session between Positions 3 and 6 in the serial presentation. There were 240 trials in total, split into three blocks. Each rest period between blocks was dynamically adjusted to ensure the task duration (time spent training) was matched to the trained-on single/dual task. The presentation speed was dynamically adjusted to a threshold for each participant so that accuracy remained at ∼70%. In the pretraining session, each stimulus was initially presented with a 100 ms duration and then scaled down in 10 ms increments if participants accurately responded to 5 trials in a row. If two incorrect trials occurred in succession, the presentation duration would increase by 10 ms per stimulus. The minimum stimulus duration was 10 ms. Each subsequent session started with the final presentation duration (i.e., 20 ms) of the pretraining session. Presentation duration is the dependent variable with shorter durations indicating better performance. Critically, accuracy and presentation duration are the main performance measures. Each 240-trial session lasted ∼13 min. This task was also used in the pre-, post-, and follow-up testing sessions as a transfer task for the four groups that trained on training multitask.

Multitask training transfer

To assess "near" transfer (i.e., transfer to a highly similar task to that trained), we used the same task structure as the training multitask but used different visual stimuli ("%," "#," and "&," subtending a visual angle of approximately 1.2°) and different complex tones (Dux et al., 2006).

Visual search

We used a classic inefficient visual search task (find a "T" among distractor "L's") that taps spatial attention mechanisms as our main transfer task as we have found transfer to this domain in two previous experiments that combined tDCS with cognitive training (Filmer et al., 2017a,b). This task is considered a far transfer task, for in contrast to both training tasks, it requires a different response to different stimuli (finding a stimulus among similar distractors vs selecting the correct response to a presented stimulus) and thus is primarily taxing a distinct cognitive domain. Trials began with a fixation dot of 0.25° diameter, which lasted for a randomly determined duration of either 0.4 or 0.6 s. Following this the fixation cross would disappear and an array of "L" stimuli plus a single target "T" stimulus appeared (total array subtended 17° of visual angle). These stimuli appeared equidistant from one another in random positions within the search array. Participants were required to find the "T" rotated either 90° or 270°—among randomly rotated—distractor letter "L's." Making the distractors similar to the target makes the search less efficient and the task more difficult (Wolfe and Horowitz, 2004). These two stimuli had an approximate linewidth subtending 0.2° and a height of 0.8° of visual angle. Task difficulty/search time was manipulated by varying the number of distractor "L's," either 7, 11, or 15, with 80 trials for each task difficulty, totaling 240 trials. Participants had 3,000 ms to use their left index finger to press the "Z" key if the "T" was rotated to 270° or their right index finger to press the "M" key if the "T" was rotated to 90°. Practice consisted of 15 trials before each of the pre-, post-, and follow-up testing sessions; feedback was given for incorrect responses during the practice in the form of a short auditory tone.

Random dot motion

Participants were presented with a random dot kinematogram, which consisted of two overlapping sets of dots [160 black (RGB: 0 0 0) and white dots (RGB: 255 255 255), respectively] simultaneously moving in different (incongruent) or similar (congruent) directions. Each dot was set to be 1 pixel in size, with the total dot display area (circular) set to 400 pixels in diameter (11°). The participants were cued to track relevant dots of a particular color prior to the commencement of a block of trials. They were asked to indicate whether the target-colored dots were moving to the left or right of the vertex, as accurately and quickly as they could. The angular offsets were varied to create differences in task difficulty (15° 75°; 15° −45°; 30° 90°; 30° −30°; −15° 45°; −15° −75°; −30° 30°; −30° −90°). During each trial, participants were instructed to keep their eyes centrally fixated on a red fixation dot (subtending approximately 0.2° of visual angle). Practice consisted of 10 trials with each cued color, before each of the pre-, post-, and follow-up testing sessions. During the test phase, participants completed 80 trials with each cued color, with 10 trials at each of the eight different angular offset configurations. Performance was recorded as the average accuracy and reaction time for each of the angular offset configurations.

Go/no-go with mind-wandering

Previous work from our lab has highlighted that tDCS can induce higher self-reported rates of mind-wandering during training with higher intensities of stimulation, that is, 2.0 mA (Filmer et al., 2019b). To assess mind-wandering here, participants responded to a probe that appeared pseudorandomly (every 15–25 trials) throughout a simple go/no-go task. The probe consisted of a question asking "To what extent have you experienced task-unrelated thoughts prior to the thought probe? 1 (minimal)–7 (maximal)." Participants had to respond within a 10 s window. Upon responding, the go/no-go task resumed. A total of 5 probes appeared per block during the go/no-go task (4 blocks in total). For the go/no-go task, participants were familiarized with two similar, yet distinct visual stimuli (a spiky and a smooth white abstract 3D shape, both subtending approximately 1.2°). The first was the "go" stimulus, which was pseudorandomized so that half of the participants had the spiky object, and the other half the smooth object, as the "go" stimulus. The second stimulus, the "no-go" stimulus, was whichever object they didn't have as the "go" stimulus. Trials began with a fixation cross of 0.5° diameter and 0.1° linewidth, which lasted for a randomly determined duration of either 0.2, 0.3, 0.4, 0.5, or 0.6 s. Following these one of the two stimuli would appear and responses were given by keypress each time the go stimulus appeared (75% of trials), while withholding their response to the no-go stimulus (25% of trials). Participants were instructed to respond as quickly as possible. The main measure of interest for this task is the stopping accuracy on the no-go trials where participants needed to withhold their response.

Operation span

We used an automated version of the operation-span working memory task (O-Span; Unsworth et al., 2005). This paradigm includes visual stimuli (letters and numbers) and involves retaining information while undertaking a different task. Specifically, a fixation square of 0.4° diameter, which lasted for a randomly determined duration of either 0.2, 0.3, 0.4, 0.5, or 0.6 s, was followed by a simple math equation (i.e., "$(2 \times 4) - 3 = ?$"), which was centrally presented and required a mouse click using a standard computer mouse to indicate they had solved the problem. Next, a solution to the equation (i.e., "5") was centrally presented where participants had to indicate whether the presented answer was TRUE or FALSE, via a mouse click. After the response to the math equation, a single letter (either of the following: "F," "H," "J," "K," "L," "N," "P," "Q," "R," "S," "T," or "Y," subtending approximately 0.8°) was presented for 800 ms, which participants had to hold in memory. Another trial beginning of the same structure would immediately begin following the letter presentation. Once the block of trials was finished, a response screen displayed all possible letters in a grid array. Participants had to use a mouse to click on the letters that they had seen throughout the block in the correct presentation order. The number of trials per block was randomized to modify difficulty (3–7 trials per block) for a total of 15 blocks.

A partial credit unit score was used to assess performance, where the fraction of correctly reported letters for each block was summed across blocks and divided by the total number of blocks to give a participant's score. This task was considered a far transfer task as it required performing simple math operations and retaining information in the face of distraction, responding with a mouse- rather than a keypress, and having a focus on accuracy rather than speed, clearly differentiating it from the trained multitask.

Digit span (forward and reverse)

Forward. Participants were presented with a stream of digits and were asked to recall them in order. Each correct response led to a trial with an additional digit to recall. Two consecutive trials containing errors lead to a trial with one less digit. There was a total of 15 trials. A partial credit unit score was used to assess performance, where the fraction of correctly reported numbers for each trial was summed across trials and divided by 15 to give a participant's score.

Reverse. After completing the forward component of the task as described above, participants were presented with a stream of digits and were asked to recall them in reverse order. Each correct response led to a trial with an additional digit to recall. Two consecutive trials containing errors lead to a trial with one less digit. There were also a total of 15 trials, and performance was again assessed by the participant's partial credit unit score.

Dynamic dual task (DDT)

We used a continuous visuomotor tracking task combined with a perceptual discrimination task (adopted from Bender et al., 2017) to further explore multitasking ability. This task combined rapid perceptual decisions (reaction time based) with a continuous tracking task (accuracy based). This differs from the other single/dual multitask paradigms, which primarily assess improvements in reaction times whereas DDT's main dependent variable is accuracy. During the tracking component of the task, participants moved a computer mouse with the goal of keeping the cursor (1.2°) inside of a moving disc (5.5° diameter) on the screen while ignoring the centrally presented shapes. The trajectory of the disc was generated by the visible target bouncing off the display boundaries or two invisible discs in a Newtonian direction. Movements in the *x*-axis and *y*-axis were calculated independently and to ensure that the tracking target did not overlap with the centrally presented shapes, a 1 cm protective radius was implemented around the shapes. Accuracy was recorded as the percentage of time that the cursor was kept within the edges of the disc. During the discrimination component of the task, participants were cued to a particular colored shape before the trial began. They then responded with the spacebar as quickly as possible when the target shape was presented in the center of the screen and not when a distractor was presented while ignoring the moving disc. There were 12 possible colored shapes (red, green, blue, or yellow) represented as either a square, hexagon, or star, with an approximate size of 4.5°, with the distractor shapes occurring at a 50% frequency. The shapes were presented every 2, 2.5, or 3 s during a trial. During the multitasking component of the task, participants performed both the tracking component and the discrimination component simultaneously. Multitasking performance was estimated by calculating the difference in accuracy between the single-task trials and the dual-task trials for both the tracking and discrimination components.

Both the speed of the moving target disc and the response window duration for the target shape were manipulated with a thresholding procedure before the task began in the first session to ensure ∼80% accuracy on each single-task condition. Specific details of this thresholding procedure can be found in Bender et al. (2017). In short, the tracking target speed to be used throughout the experiment was determined from 9×60 s tracking trials, where the disc's speed would change (between 0.01 and 0.112 degrees per second) to maintain 80% tracking accuracy. The response window duration for the shape discrimination component was determined from 9×20 s trials where the response window would change between 250 and 1,000 ms to maintain 80% discrimination accuracy. A centrally presented fixation square (0.5°) was presented during the thresholding session. The square would turn green for 50 ms if the target shape was accurately responded to within the current response window or when a response to a nontarget shape was successfully inhibited. The square would turn red for 50 ms if the target shape was not responded to within the allotted response window or if a nontarget shape was responded to.

Imaging acquisitions

As previously described (Rideaux et al., 2022), MRS was conducted on a MAGNETOM 7T Plus MRI scanner (Siemens Healthcare) equipped with a 1-channel transmit and 32-channel receive head coil (Nova Medical Centers). Anatomical T_1-weighted images were acquired for voxel placement with an MP2RAGE sequence: TR = 4,300 ms, TE = 3.38 ms. We opted for a combination of MEGA-semi-LASER (MEGA-sLASER) for GABA quantification and semi-LASER for glutamate due to the distinct advantages offered by these specific sequences. Although it is possible to detect both GABA and glutamate from non-editing approaches at 7T, the MEGA editing technique provides superior specificity for GABA by effectively separating its signal from overlapping metabolites. This specificity is essential given GABA's relatively low concentration in the brain, which can make it challenging to accurately measure without MEGA editing. In addition, the semi-LASER sequence allowed us to reliably quantify glutamate without the need for editing, thus maximizing our data acquisition efficiency and overall experimental robustness. For GABA+ measurements, spectra were acquired using a research MEGA-sLASER sequence (Andreychenko et al., 2012) with parameters: TE = 74 ms, TR = 7,800 ms, and bandwidth = 4,000 Hz; 64 transients (32 ON and 32 OFF) in a total of 4,096 data points; and a 23.6 ms Gaussian editing pulse applied at 1.90 (ON) and 7.46 (OFF) ppm. Water reference spectra (two transients) were acquired without water suppression. Water suppression was achieved using variable power with optimized relaxation delays (VAPOR; Tkáć and Gruetter, 2005), and fat suppression was achieved with outer volume suppression. For the detection of glutamate, spectra were acquired using a research sLASER sequence (Scheenen et al., 2008) with the following parameters: TE = 42 ms, TR = 7,790 ms, spectral width = 4,000 Hz; 32 transients of 4,096 data points. Additionally, in line with recent expert consensus (Cudalbu et al., 2021), a single inversion recovery metabolite nulling in the sLASER (TI = 1,135 ms) was performed in two subjects to measure macromolecule contribution; these results were used to constrain metabolite quantification. Automated shimming with 3D GRE brain, followed by a research FAST(EST)MAP shimming sequence, was run in succession. All MRS data were analyzed from their Siemens TWIX format. During the MRS acquisition, participants passively viewed a naturalistic film. The voxel order was counterbalanced across participants (half of the participants had the prefrontal voxel measured first). The acquisition window for the MRS sequences was 1,000 ms; however, there was no signal in the free induction decay after 250 ms, so we only used the first 250 ms of the data for each acquisition. Spectra were acquired from voxels ($40 \times 26 \times 26$ mm^3) targeting frontal and occipital cortices (Fig. 5*A*). The frontal voxel was positioned on a slice 1.5 mm above

the superior margin of the lateral ventricles, with the center of the volume placed one-third of the total anterior/posterior distance, centered halfway between the left lateral border and the midline. The occipital voxel was positioned centrally across the bilateral midline, above and aligned to the calcarine fissure. The coordinates of the voxel locations were used to draw a mask on the anatomical T_1-weighted image to calculate the fractional volume of gray matter, white matter, and cerebrospinal fluid within each voxel. Tissue segmentation was performed using the Statistical Parametric Mapping toolbox for MATLAB (SPM12, www.fil.ion.ucl.ac.uk/spm; Ashburner and Friston, 2005).

Data processing

Spectral preprocessing and quantification were conducted in MATLAB using in-house scripts as described below. Consistent with previous work (Rideaux et al., 2019, 2022; Rideaux, 2020, 2021), to improve frequency/phase estimation, all edited (MEGA) subspectra were 3 Hz exponentially line-broadened and zero-filled to a spectral resolution of 0.061 Hz/point. Unedited (sLASER) data were not line-broadened or zero-filled due to constraints imposed by LCModel (Provencher, 1993). For all data, frequency and phase parameter estimates were obtained by modeling the total creatine (creatine and phosphocreatine) signal, and then these parameters were used to align subspectra to a common frequency and phase, as previous work has shown this is the most effective method of alignment for GABA+ quantification (Rideaux et al., 2021). Estimates of the signal area and full width at half maximum (FWHM) were also obtained, and subspectra (and their corresponding ON/OFF subspectra) with parameter estimates >2.5 standard deviations (SD) from the mean within a scan were omitted from further analysis.

For the edited data, the average ON and OFF spectra were subtracted to produce the edited spectrum (Fig. 5*B*), from which GABA+ (3 ppm) signal intensity was modeled using a Lorentzian. Similarly, water signal intensity was determined by fitting a Lorentzian model to the average water-unsuppressed spectra at 4.7 ppm. For the unedited data, glutamate and water signal intensity were quantified using the LCModel. The macromolecule spectrum was included to constrain glutamate quantification. The LCModel basis set consists of density-matrix simulated basis spectra of 16 metabolites in addition to an experimentally acquired macromolecule spectrum: alanine, aspartate, creatine, GABA, glucose, glutamine, glutamate, glutathione, glycerophosphocholine, lactate, myo-inositol, N-acetylaspartate, N-acetylaspartylglutamate, phosphocreatine, phosphoethanolamine, and taurine.

GABA and glutamate concentrations were calculated as the area of the fitted peaks and expressed in institutional units using the unsuppressed water signal as an internal concentration reference. That is, we applied a tissue-correction method that accounts for differences in relaxation times across the tissue types within a voxel (Gasparovic et al., 2006). The assumed longitudinal relaxation times of water, GABA+, and glutamate at 7T were 1.78 s, 1.33 s, and 1.68 s, respectively (Andreychenko et al., 2012; Xin et al., 2013). Their assumed transverse relaxation times were 0.05 s, 0.09 s, and 0.09 s, respectively (Andreychenko et al., 2012; Marjańska et al., 2012). The assumed relative densities of MR-visible water in gray matter, white matter, and cerebral spinal fluid were 0.78, 0.65, and 0.97, respectively (Ernst et al., 1993).

During data collection, there was an update to the 7T scanner, which enabled automatic subject-specific B_1 calibration, whereas before the upgrade we used a constant reference voltage of 275 V to facilitate faster workflow (appropriate for an average subject). We found that this change resulted in different average metabolite values before and after the upgrade. Thus, to control for the associated changes, all metabolite values were first corrected for the scanner version prior to analyses. Data were corrected by using a linear mixed effects model to model the variance associated with updating the scanner and using the residual values for all subsequent analyses. For data quality of the nonedited spectra, FWHM, the signal-to-noise ratio (S/N), and the Cramér–Rao lower bound (CRLB) were assessed based on the LCModel output. The FWHM values were converted from parts per million (ppm) to Hertz (Hz) using the Larmor equation, with a field strength of 7T and a gyromagnetic ratio of 42.577 MHz/T for protons. For the frontal cortex voxel: FWHM = 7.75–13.71 Hz, mean = 10.60 Hz, SD = 1.54 Hz; S/N = 23–88, mean = 64.55, SD = 10.83; CRLB = 3–9, mean = 3.99, SD = 1.03. For the visual cortex voxel: FWHM = 7.75–13.71 Hz, mean = 9.77 Hz, SD = 1.47 Hz; S/N = 22–85, mean = 68.49, SD = 8.93; CRLB = 3–15, mean = 3.74, SD = 1.12.

Statistical analysis

We employed Bayesian methods to evaluate the strength of evidence toward or against the null and alternative hypotheses regarding task performance in the cognitive battery pre- and post-training. Corrections were made for multiple comparisons where applicable, for as noted above a Bayesian approach makes more conservative comparisons compared with null hypothesis testing procedures, in that intervals for comparisons are more likely to include zero and therefore more likely to be valid (Gelman and Tuerlinckx, 2000). Across our control groups and the performed tasks, it was expected that we would find some tasks and groups that show no effect from training, so we wanted to be able to assess the strength of evidence for these null effects. BF_{10} of >3 were taken to demonstrate meaningful evidence for the alternative hypothesis, whereas a $BF_{10} < 0.333$ meant meaningful evidence for the null hypothesis (Wagenmakers et al., 2018; van Doorn et al., 2020). The following procedures were determined before performing our key hypothesis tests. To ensure sufficient trial numbers for analysis and to confirm proper task comprehension and completion, for all tasks where reaction times were the focal performance metric (training multitask, transfer multitask, dynamic dual task, go/no-go, and visual search), we excluded participants from a particular task's analyses if their mean accuracy for either the pre- or -post-training session fell below 60%. This is in line with our previous studies' protocols.

Following this, at the individual participant level, trials were subjected to an outlier removal analysis by calculating the interquartile range (IQR) for the reaction time distributions, multiplying this range by 1.5, and adding this value to the third quartile (Q3 + 1.5 × IQR). For each participant, reaction times that were greater than this threshold were excluded. This was calculated within the specific trial type for tasks where multiple stimulus types were presented. For example, for the visual search task, 1.5 × IQR was calculated separately for the reaction times from the three different set sizes. Similarly, for the multitasks, 1.5 × IQR was calculated for the single-auditory, single-visual, dual-auditory, and dual-visual trials separately. This outlier exclusion approach was selected due to superior outlier detection compared with using the >3 SD approach. This was assessed by visualizing a random subset of 10 participants' reaction time data distributions with a histogram and comparing the distributions before and after outlier exclusion under the two approaches. As reaction time data rarely conforms to a Gaussian distribution, the IQR approach assumes less about the data in assessing outliers (Whelan, 2008). Using the IQR approach, slightly more of the trials at the extreme right side of the distribution (i.e., unusually slow reaction times) were excluded from subsequent analyses. Reaction times faster than 0.2 s were also excluded, in line with common practice.

We also applied the same Q3 + 1.5 × IQR exclusion threshold at the participant level for each task, where a mean reaction time exceeding this threshold for either the pre- or post-training sessions resulted in that participant being excluded for analysis on the respective task. Our preregistration stated that we would use an exclusion threshold >3 SD for the trained multitask. For the sake of consistency with the other tasks and with the trial-level approach, we have reported data that used the IQR outlier analysis for the multitask as well. To evaluate the effect of combined training and tDCS on the trained multitask, we used a Bayesian *t* test on the change in performance (single- and dual-task reaction time and accuracy) between the pre- and post-training session and the pre- and ∼30 d follow-up session. This approach was employed as we had specific hypotheses for the double-blind groups and for the active controls. For all transfer tasks, the respective key performance metrics were compared between the pre- and post-training session and the pre- and ∼30 d follow-up session to evaluate the presence of training transfer. Bayesian correlations were run on individual group reaction times for each task, and key metabolites (glutamate, GABA+, and the ratio of glutamate/GABA+). Fisher *z* transformations were used to compare correlation coefficients between groups.

Results

Training improves task-specific performance

We contrasted the single-task RSVP training with a single/dual multitask training paradigm due to their divergent cognitive demands: while they both tap information processing limits, the former is predominately perceptual whereas the latter recruits more executive operations (Marois and Ivanoff, 2005). Given that specific training typically augments task performance, we expected that individuals in the RSVP control training group (who did not train on the multitask) would exhibit poorer multitask performance after training relative to their multitask training counterparts.

The RSVP training group displayed less single- and dual-task improvements relative to the other multitasking training groups from pre- to post-training. Using independent *t* tests, we compared each group to the RSVP training group. We found moderate to strong evidence for single-task improvements in the sham group, and 1 and 2 mA left PFC stimulation and multitasking groups (sham, $BF_{10} = 4.588$, $p_{\text{-bonf}} = 0.040$; 1.0 mA left PFC, $BF_{10} = 19.123$, $p_{\text{-bonf}} = 0.007$; 2.0 mA left PFC, $BF_{10} = 3.558$, $p_{\text{-bonf}} = 0.055$). However, the evidence for the 1.0 mA right PFC and multitasking group was anecdotal ($BF_{10} = 2.315$, $p_{\text{-bonf}} = 0.095$). Similarly, dual-task improvements were pronounced in the sham and both left PFC stimulation groups, compared with the RSVP group (sham, $BF_{10} = 1{,}739.306$, $p_{\text{-bonf}} < 0.001$; 1.0 mA left PFC MT, $BF_{10} = 156.638$, $p_{\text{-bonf}} = 0.001$; 2.0 mA left PFC MT, $BF_{10} = 376.852$, $p_{\text{-bonf}} < 0.001$), whereas the 1.0 mA right PFC group showed only anecdotal improvement ($BF_{10} = 2.834$, $p_{\text{-bonf}} = 0.074$). RSVP training participants reduced the exposure duration required to accurately identify the target more than those who trained on the multitask paradigm (all independent *t* test comparisons; $BF_{10} > 223$, $p_{\text{-bonf}} < 0.001$). These results were still evident 30 d after training when we considered the change in performance from pretraining to the follow-up session (all comparisons; $BF_{10} > 10$, $p_{\text{-bonf}} < 0.02$; Fig. 3).

In line with previous results, despite observable task-specific multitasking training effects, brain stimulation did not yield

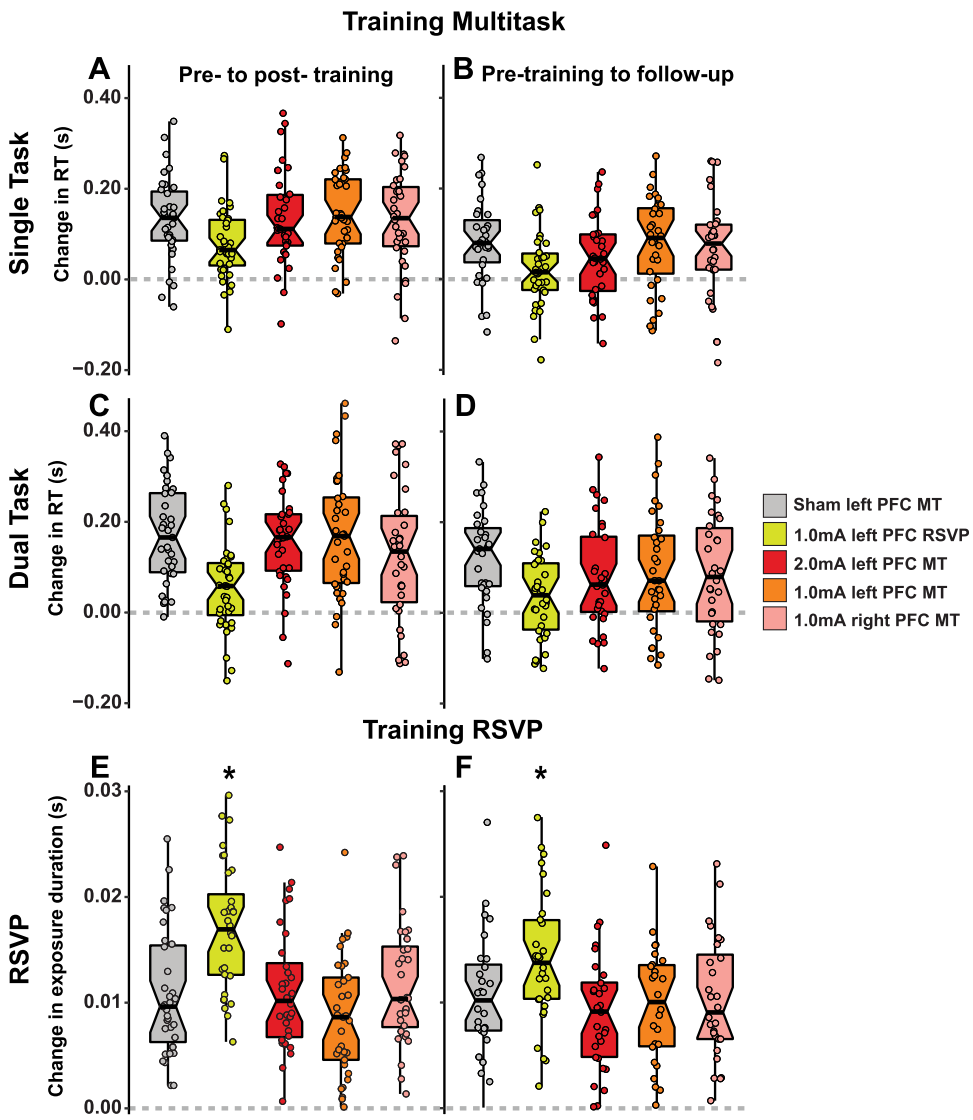

Figure 3. Training- and stimulation-induced changes in performance (***A***, ***C***) represent the change in reaction time from pre- to post-training for the training multitask, with ***A*** representing the change in single-task reaction time and ***C*** representing the change in dual-task reaction time. ***B***, ***D***, The sustained impact of training from pretraining to the 30 d follow-up post-training, with ***B*** representing the change in single-task reaction time and ***D*** representing the in dual-task reaction time. ***E***, The changes in presentation duration (s) from pre- to post-training. ***F***, The changes in exposure from pre- to 30 d follow-up. Individual dots represent each participant's change in performance (s). * denotes $BF_{10} > 3$ compared with all other groups.

additional benefits when comparing pre- with post-training or follow-up session performance. We found moderate evidence against a stimulation-induced benefit for the single-task component of the multitasking paradigm, relative to sham, for all other multitasking training groups (*t* tests on change scores: 1.0 mA left PFC, $BF_{01} = 3.641$ $p_{\text{-bonf}} = 1.000$; 1.0 mA right PFC, $BF_{01} = 4.028$ $p_{\text{-bonf}} = 1.000$; 2.0 mA left PFC, $BF_{01} = 4.035$, $p_{\text{-bonf}} = 1.000$). The same pattern was observed for the dual-task condition (1.0 mA left PFC, $BF_{01} = 4.049$ $p_{\text{-bonf}} = 1.000$; 1.0 mA right PFC, $BF_{01} = 1.613$, $p_{\text{-bonf}} = 0.564$; 2.0 mA left PFC, $BF_{01} = 3.623$, $p_{\text{-bonf}} = 1.000$). Additional analyses were conducted to compare active groups to each other. The BF_{10} indicated anecdotal to moderate evidence in support of the null hypothesis ($BF_{01} = 1.7–3.8$). These outcomes suggest that neither the site of stimulation (left PFC vs right PFC) nor the intensity of stimulation (1.0 vs 2.0 mA) had a significant effect on task performance. In comparing active groups with each other and with the sham group, no substantial differences were found. Collectively, the findings suggest that tDCS did not influence the outcome of multitasking training, regardless of its intensity or site, in line with previous findings from our group using this same experimental design (Filmer et al., 2017a).

Training paired with stimulation facilitates far transfer

Despite observing no stimulation benefits for the trained-on task (we return to this in the Discussion), our results show clear evidence for far transfer on the visual search task—an observation that aligns with previous results (Filmer et al., 2017a). Importantly, the administration of 1.0 mA stimulation targeting either the left PFC or right PFC, when combined with multitasking decision-making training, led to significant enhancement in search performance from pre- to post-training. This enhancement scaled with set size, as shown in (Fig. 4). Using two-tailed independent *t* tests, specifically, relative to sham, there were significant enhancements in search performance with 1 mA stimulation targeting either the left PFC ($BF_{10} = 5.742$, $p_{\text{-bonf}} = 0.030$ for Set Size 12; $BF_{10} = 6.643$, $p_{\text{-bonf}} = 0.025$ for Set Size 16) or right the PFC ($BF_{10} = 14.915$, $p_{\text{-bonf}} = 0.009$ for Set Size 12; $BF_{10} = 122.788$, $p_{\text{-bonf}} = 0.001$ for Set Size 16). Interestingly, the left PFC effect was dosage dependent: 2 mA stimulation showed minimal transfer effects relative to sham when compared with sham across both pre- to postsessions ($BF_{10} = 1.073$, $p_{\text{-bonf}} = 0.269$ for Set Size 12; $BF_{10} = 0.962$, $p_{\text{-bonf}} = 0.313$ for Set Size 16). Conversely, the 1 mA RSVP training group, which was subject to the same stimulation intensity but with the control task training, demonstrated indeterminate to anecdotal transfer effects when compared with sham at pre- to postsessions (Set Size 12, $BF_{10} = 1.956$, $p_{\text{-bonf}} = 0.121$; Set Size 16, $BF_{10} = 1.006$, $p_{\text{-bonf}} = 0.296$). Finally, when comparisons were made between the active groups, no compelling differences emerged. Our analyses yielded anecdotal to moderate evidence (BF_{01} range, 2.4–3.8) favoring the null hypothesis, suggesting that neither the site of stimulation (left PFC vs right PFC) nor the dosage of stimulation (1 vs 2 mA) significantly differentiated the impact on visual search performance when each active group was compared with each other.

Training transfer is sustained

Additionally, we evaluated the sustainability of training transfer effects. Approximately 30 d post-training, participants returned to complete a final cognitive battery (the follow-up session). We observed sustained transfer in the visual search task, with benefits from combined training and stimulation persisting for both left and right PFC stimulation when compared with sham at Set Size 16 (left PFC, $BF_{10} = 6.080$, $p_{\text{-bonf}} = 0.029$; right PFC, $BF_{10} = 54.284$, $p_{\text{-bonf}} = 0.002$). However, we found limited evidence of transfer to either Set Size 8 or 12 in this follow-up session (for all conditions, $0.333 < BF_{10} < 3$, all $p_{\text{-bonf}} > 0.05$). Moreover, limited evidence of a performance change was detected between the post-training session and the follow-up (for all conditions, $0.333 < BF_{10} < 3$, all $p_{\text{-bonf}} > 0.05$). Thus, our protocol yields prolonged training effects on spatial attention that persist following multitasking training, combined with both left and right hemisphere tDCS. Further group comparisons are detailed in Table 2 below.

Neurochemistry predicts transfer

We examined the potential role of key neurotransmitters in the context of stimulation-induced transfer. Normal brain function relies on a balance between neural excitation and inhibition, with glutamate and GABA playing crucial roles in maintaining this balance (Shadlen and Newsome, 1994; van Vreeswijk and Sompolinsky, 1996; Petroff, 2002; Vogels and Abbott, 2009; Marín, 2012). Altered activities of these neurotransmitters have been found to underpin efficient information transfer and neuroplastic changes associated with learning (Anderson et al., 2000; Wehr and Zador, 2003; Wilent and Contreras, 2005; Poo and Isaacson, 2009). We hypothesized that tDCS—which is thought to modulate cortical excitability via its influence on neurotransmitters—might interact with these mechanisms to impact training outcomes. We used ultrahigh field 7T MRS to measure baseline concentrations of glutamate (Glu) and GABA+ (comprising

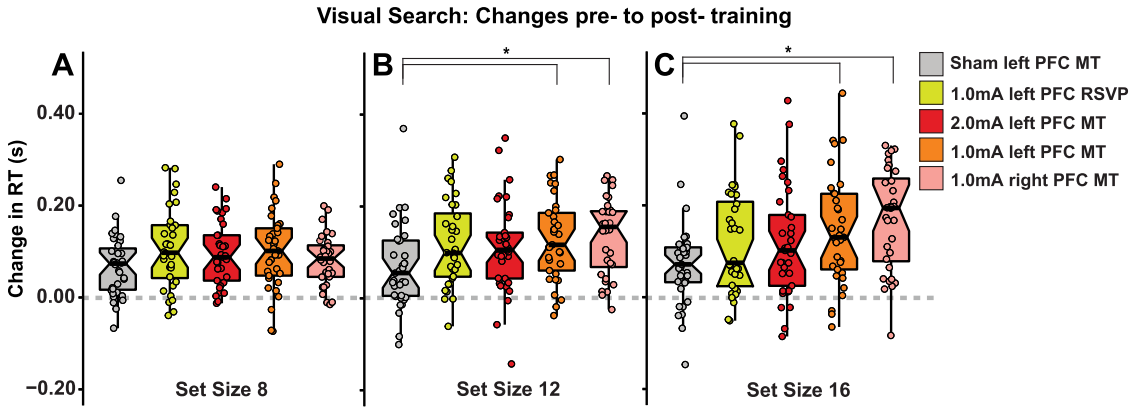

Figure 4. Training transfer to visual search across set sizes. Change in reaction time from pre- to post-training for the transfer visual search task for (***A***) Set Size 8, (***B***) Set Size 12, and (***C***) Set Size 16. Individual dots represent each participant's change in performance (s) for each group.

GABA and macromolecules). There was substantial evidence that the relative concentration of glutamate in the left frontal region at baseline correlated with the change in reaction time for Set Size 16 on the visual search task, exclusive to the 1.0 mA right PFC stimulation group ($BF_{10} = 6.156$, $r = 0.486$; all others, $0.264 < BF_{10} < 0.943$), the group that showed the largest training plus stimulation transfer effect. Using a Fisher z test of correlation coefficients, we found a significant difference between sham and the 1.0 mA right PFC correlations ($z = 2.9318$, $p = 0.0034$). Critically, these results are specific to neurochemical glutamate concentrations in the frontal cortex, as the same analyses demonstrated there were no reliable correlations between search performance and concentrations of glutamate in the visual cortex, with moderate evidence against a relationship for sham, 1.0 mA right PFC and 2.0 mA left PFC ($BF_{01} > 3.571$), anecdotal evidence for the 1.0 mA RSVP group ($BF_{10} = 2.254$), and against in the 1.0 mA left PFC ($BF_{01} = 2.012$). Furthermore, there was no difference between using water (reported values) or creatine as an internal reference. Previously, we had found that the ratio of GABA and glutamate [excitatory/inhibitory (E/I) balance] was related to the magnitude of training disruption in response to cathodal stimulation in the PFC for a different training task protocol (Filmer et al., 2019a). However, here we do not find any relationship between GABA or the *E/I* ratio and transfer to any task (Fig. 5).

Table 2. Group comparisons for the inefficient visual search task

Group comparison		Set Size 8		Set Size 12		Set Size 16	
		BF_{10}	$p_{\text{-bonf}}$	BF_{10}	$p_{\text{-bonf}}$	BF_{10}	$p_{\text{-bonf}}$
		Change in reaction time from pre- to post-training					
Sham vs	1.0 mA LPFC RSVP	1.505	0.171	1.956	0.121	1.006	0.296
	2.0 mA LPFC MT	0.799	0.407	1.075	0.269	0.962	0.313
	1.0 mA LPFC MT	0.926	0.328	**5.742**	**0.030**	**6.643**	**0.025**
	1.0 mA RPFC MT	0.359	1.000	**14.915**	**0.009**	**122.788**	**0.001**
		Change in reaction time from pretraining to follow-up					
Sham vs	1.0 mA LPFC RSVP	0.382	1.000	0.504	0.876	0.569	0.721
	2.0 mA LPFC MT	0.501	0.882	0.450	1.000	1.591	0.165
	1.0 mA LPFC MT	0.685	0.539	1.242	0.230	**6.080**	**0.029**
	1.0 mA RPFC MT	0.466	1.000	0.541	0.797	**54.284**	**0.002**

BF_{10} denotes $p(H_1)/p(H_0)$, where the alternate hypothesis (H_1) is that there is a difference in the change of reaction times from the pretraining session, to either the post-training or ~30 d follow-up; BF_{10} 3–10 indicate moderate evidence for H_1, 10–30 indicate strong evidence for H_1, 30–100 indicate very strong evidence for H_1, and >100 indicate extreme evidence for H_1. *p*-values were adjusted using Bonferroni's correction for multiple comparisons accounting for a family of four group comparisons. The bold values indicate Bayes Factors greater than 3 and *p*-values less than 0.05. Both can be considered significant, although Bayes Factors represent the likelihood of the results given the data, while *p*-values represent the probability of obtaining the observed results, assuming the null hypothesis is true.

Baseline differences in performance do not underlie training and stimulation efficacy

Baseline differences can be problematic when assessing change in performance in interventional studies. To mitigate potential baseline confounds, we used an algorithm to pseudorandomize the allocation of participants to groups (Ehrhardt et al., 2021) based on the key characteristics of age, sex, time of intervention (am or pm), multitask performance at baseline (single- and dual-task reaction times), and the order in which participants completed the tasks in the baseline battery. Using one-way Bayesian ANOVAs, we found evidence against baseline differences between all groups for all tasks ($BF_{01} > 7.8$ for all).

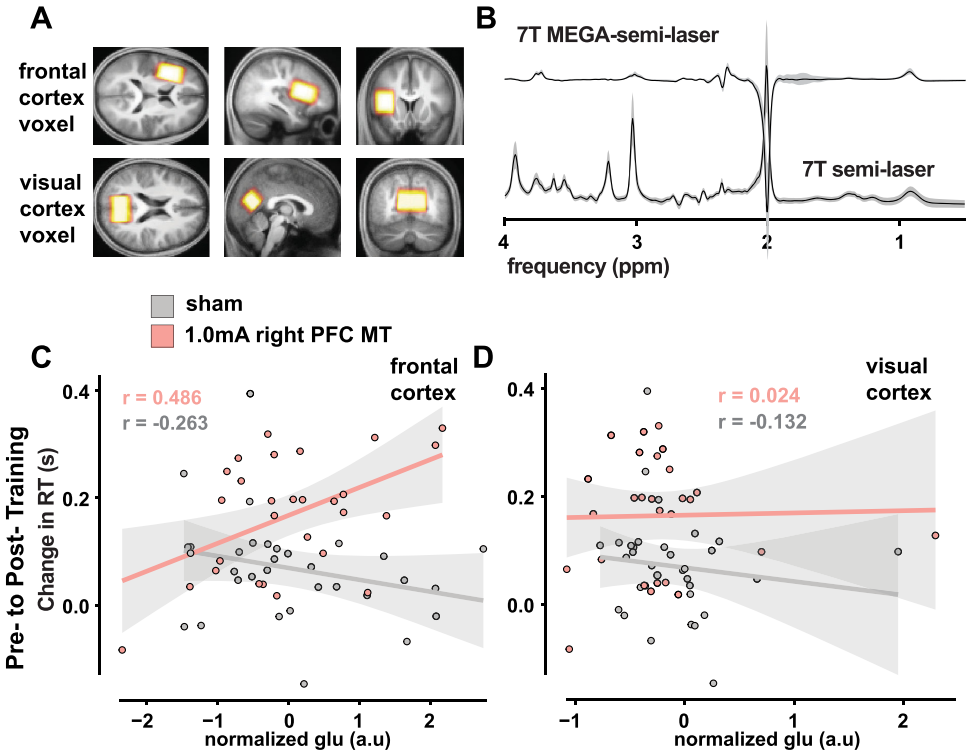

Figure 5. Neurochemistry predicts visual search transfer. ***A***, Frontal cortex and visual cortex MRS voxels shown in axial, sagittal, and coronal views placed on the average T_1-weighted structural image in MNI space. Color intensity indicates voxel placement overlap across participants. Zero is the minimum overlap between voxels, that is, there are no voxels in that location, and 1 is the maximum, that is, 100% of the voxels were in that location. ***B***, The combined semi-LASER (32 averages) and MEGA-sLASER (64 averages) spectra, acquired from the frontal cortex voxel, for all participants. The shaded regions indicate ±1 standard deviation. ***C***, Correlation between normalized glutamate concentration and the change in reaction time for Set Size 16 in the visual search task from pre- to post-training ($N = 60$, both groups combined) in the frontal cortex. ***D***, Correlation between normalized glutamate and change in reaction time for Set Size 16 in the visual search task from pre- to post-training in the visual cortex. Note: the outlier analysis was conducted using the whole sample. The two points on the far right in panel ***D*** lie within range.

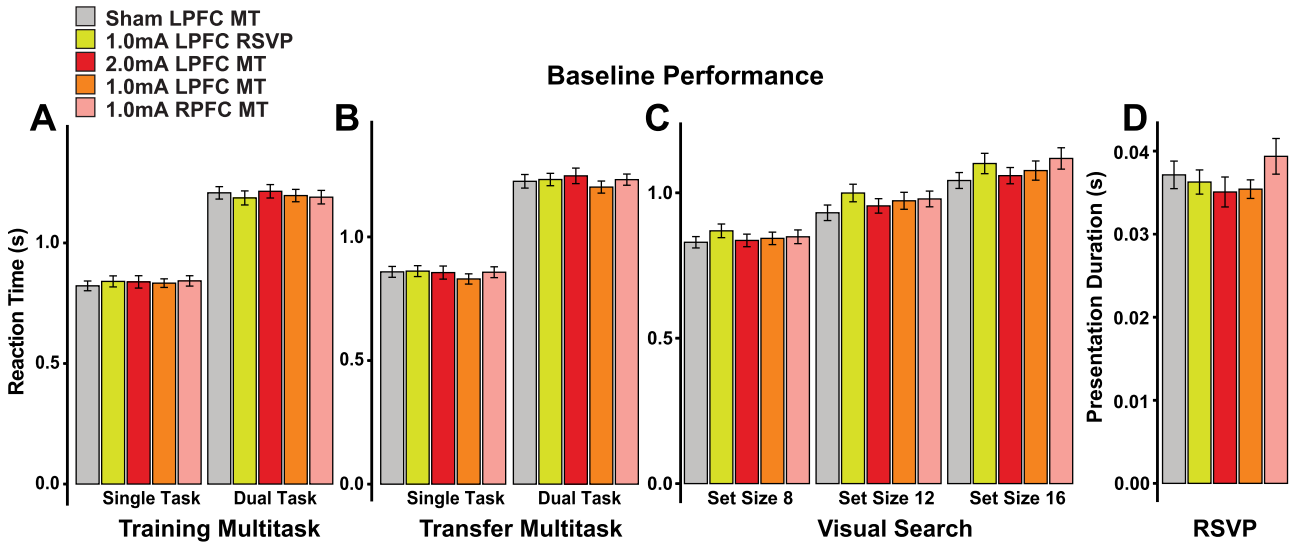

Figure 6. Baseline performance measures for key tasks (***A–C***) depict the baseline reaction times for the training multitask, transfer multitask, and visual search task, respectively. These are pivotal baseline measures ensuring comparability across groups. ***D***, The average presentation duration for the control training RSVP task, an additional key baseline measure. Error bars indicate SEM, reflecting the variation within each group. The absence of significant differences between groups at baseline validates the comparability of groups prior to treatment implementation. s, seconds.

Thus, we can conclude the algorithm successfully minimized discrepancies between groups at baseline (Fig. 6).

Stimulation efficacy is not related to a failure of blinding

A critical manipulation check in noninvasive brain stimulation studies is to test the effectiveness of blinding. Failure to do so can impact the ability to make inferences as results may otherwise reflect peripheral effects. Specifically, sensations often associated with stimulation are itching and tingling on the scalp beneath the electrodes. Adequate care is needed when applying the electrodes to avoid related confounds.

After completing the post-training cognitive session, we asked participants to guess whether they received active or sham stimulation. Overall, there was a tendency for all groups to choose "active." Out of the 178 participants, 119 (66.8%) correctly guessed that they received either active or sham stimulation. Only 33 of the 178 participants guessed that they received sham stimulation, but only 5 (14.9%) of those guesses were correct. Thirty-one participants of the 36 in the sham group (86.1%) incorrectly guessed that they had active tDCS when they in fact had sham. In addition, the sham and 1.0 mA left PFC groups were double-blinded (neither the participants nor experimenters knew the group membership), which strongly suggests that the sham protocol was effective at blinding participants in that group to their stimulation condition. All four active stimulation groups had similar rates of correct guesses (1.0 mA left PFC = 80%, 1.0 mA right PFC = 74.3%, 1.0 mA left PFC RSVP = 86.1%, 2.0 mA left PFC = 80.6%).

We also asked participants who guessed they had active stimulation whether they thought that they had received a low (1.0 mA) or high (2.0 mA) intensity of stimulation. Overall, 52.4% of participants correctly guessed the stimulation intensity that they received (1.0 or 2.0 mA), which is not different from chance. Of participants who correctly guessed that they were in the active condition (i.e., excluding participants from the sham group who guessed that they were in the active condition), 66.6% correctly guessed the intensity they received. All active stimulation groups that received the 1.0 mA intensity had similar rates of correct guesses regarding the stimulation intensity (1.0 mA left PFC = 75%, 1.0 mA right PFC = 80.8%, 1.0 mA left PFC RSVP = 74.2%). However, the 2.0 mA group guessed at less than chance accuracy, achieving only a 37.9% correct guess rate. Together, these results suggest that stimulation location, intensity, and the task trained on, did not have an impact on participants' ability to guess which stimulation condition they received.

Transfer is domain specific

Broadly testing the extent of transfer is critical to determine the underlying mechanisms driving such effects. To test if the far transfer effects noted above extend to other cognitive domains, participants completed a range of tasks (positive and negative control tasks) assessing different cognitive domains including working memory (forward and reverse span, and an operational span task), response inhibition (go/no-go), information processing speed (RSVP), multitasking (single/dual task; 0 ms stimulus onset asynchrony), dynamic multitasking (shape discrimination coupled with a visuomotor tracking task), and selective attention (random dot motion). We found limited evidence for an influence of stimulation and/or training in any comparison for this broad range of tasks when assessing average change scores between groups (for all conditions, $BF_{10} < 3$). There were also no baseline differences between any groups for any of the tasks (for all conditions, $BF_{10} < 3$). These findings indicate that though multitasking training and 1.0 mA PFC stimulation facilitate far transfer to spatial attention, this transfer was "narrow" in scope.

Discussion

Investigating the transferability of cognitive training and brain stimulation effects is critical, as it offers the potential to expand the impact of targeted interventions over a wider array of cognitive domains. Here, we assessed if a combined multitasking training and tDCS protocol could induce transfer. In addition, given evidence for individual differences in responses to tDCS (Krause and Cohen Kadosh, 2014; Katz et al., 2017a,b; Filmer et al., 2019a; Vergallito et al., 2022), we also examined if concentrations of two key neurotransmitters, glutamate and GABA+, were associated the extent to which participants benefitted from training and stimulation.

Two distinct tDCS protocols facilitated transfer: applying 1 mA stimulation targeting the left PFC or 1 mA targeting the right PFC online, for four consecutive days with multitasking training, led to enhanced spatial attention performance, relative to the control group multitasking training group that received sham stimulation. Notably, these benefits persisted at follow-up testing ∼30 d later. Crucially, changes in performance from pretraining to both post- and follow-up sessions were particularly evident at larger search set sizes, when spatial attention is more taxed. These improvements at more challenging levels of the task underscore that our results are primarily associated with enhancements in cognitive processes, rather than motoric operations, which remain consistent across set sizes. Indeed, other work has shown that tDCS effects can be more pronounced as tasks increase in difficulty (Jones and Berryhill, 2012; Chua et al., 2017; Framorando et al., 2021; Vergallito et al., 2022). To assess the nature and breadth of transfer, multiple cognitive domains were also examined. We found transfer to be specific to spatial attention. An important side note is that when collapsing data across all groups, there were improvements between pre- and post-training sessions for all tasks likely representing practice effects (doing a task for the second time). Only the reverse digitspan and go/no-go task "go" reaction times and no-go commission errors were not improved.

As tDCS and other stimulation techniques have been shown to act via modulation of key neurotransmitter systems (Stagg et al., 2009; Heimrath et al., 2020), we assessed baseline measures of glutamate, GABA+, and their balance (the so-called *E*/*I* balance) in relation to transfer. Critically, the relative concentration of left PFC glutamate predicted the degree of transfer. To wit, we found that multitasking training combined with 1.0 mA stimulation led to a reduction in reaction time in the most difficult condition of the visual search task, aligning with the results of two previous experiments (Filmer et al., 2017a,b). In the right PFC stimulation group, performance changes were also found to correlate with glutamate concentrations. Collectively, the results illustrate not only that paired training and stimulation protocols can lead to far transfer but also that this is predicted by frontal cortex neurochemistry, giving insight into possible mechanisms associated with cognitive enhancement. We specifically targeted higher-level brain regions that are linked with executive processes and implicated in the multiple demand network (Duncan, 2010). Crucially this frontoparietal network extends bilaterally across both hemispheres and is integral to executive functions and complex cognitive tasks. Indeed, this collection of brain areas is activated across a wide range of cognitive tasks and this activity scales with task complexity (Duncan, 2010). Further to this, the present results suggest that excitatory processes in this network are predominately implicated in the transferable gains we see here, providing a potential marker for individually targeting training and stimulation interventions. Indeed, other work from our lab has shown functional changes within the multiple demand network can predict training transfer (Wards et al., 2023).

An important characteristic of the present study is that extensive controls were employed. There are mixed results within the training and tDCS literature, many of which can be attributed to poor experimental design (Polanía et al., 2018) as well as individual differences. We actively addressed many of the criticisms outlined by Polanía et al. (2018). Specifically, we included an active control training group; we used different electrode montages; we applied two stimulation intensities (1 and 2 mA); we preregistered and employed an established online stimulation paradigm and used multiple ROIs for the MRS experiment; we implemented double blinding for key contrasts; and we tested a very large sample size (final sample after exclusions: $n = 178$). Thus, it is unlikely the present findings are driven by a spurious influence of either training and/or stimulation or a neuroimaging artifact. In addition, when placing participants into the intervention groups and during the analyses, we used an automated algorithm to assign participants, ensuring baseline differences did not give rise to the present results. The importance of stratifying across multiple dimensions (done here across age, sex, baseline reaction times in the trained multitask, and time of day for testing) has recently been highlighted in the literature (Dougherty and Engle, 2019). We also found that participant blinding was successful and by testing a wide range of behavioral tasks we were able to assess the boundary conditions of the benefits afforded by our protocol. Finally, by employing a Bayesian analysis framework (along with traditional NHST approaches), we reduced the chance for type 1 error as such approaches make more conservative comparisons compared with classical procedures (Gelman and Tuerlinckx, 2000; we also included Bonferroni's *p*-statistics where applicable).

Earlier work from our lab, which focused on older adults and used 2 mA stimulation, reported no significant cognitive training or transfer effects induced by tDCS (Horne et al., 2021), a finding that aligns with our findings for the 2 mA stimulation group. Our results also align with previous findings, where similar protocols at 0.7 mA resulted in a transfer to the same visual search task (Filmer et al., 2017a,b). However, there is some variability in the findings reported to date. Specifically, we have previously shown—using the same trained-on multitask employed here—that combining four sessions of training with either 0.7 or 1 mA anodal stimulation applied over the left PFC was successful at improving training outcomes across both single and dual tasks assessed, but not for the visual search task (Ehrhardt et al., 2021). In contrast, no benefit from stimulation on the trained-on task was found in the current study, although transfer was present. A likely explanation is a key methodological difference in the duration of stimulation and amount of training (i.e., number of trials of training per session) between our study and that of Ehrhardt et al. (2021). Stimulation was applied for 20 min, and participants completed 420 training trials per session, whereas we administered tDCS for 13 min per session with participants training for only 240 trials per session. Indeed, the protocol reported here is the same as a previous study where the trained-on task also showed no benefit from stimulation, but transfer was evident (Filmer et al., 2017a). Thus, our findings are consistent with previously reported effects, and it appears that the duration of stimulation and/or the degree of training interacts with the type of training and stimulation benefits observed. Furthermore, the study highlights the dosage-dependent nature of tDCS effects. While 1 mA stimulation enhanced cognitive performance, 2 mA stimulation did not yield similar benefits when compared with sham, suggesting a nonlinear relationship between stimulation intensity and cognitive enhancement. Such results hint that there exists an optimal threshold of neuronal modulation, particularly within the context of higher-order executive function training interventions. Lower intensities may facilitate a favorable neurological environment for cognitive benefits, and there might be an optimal stimulation "window" for cognitive enhancement. Complex effects of stimulation dosage (both intensity and duration) have been found in the motor cortex (Batsikadze et al., 2013; Jamil et al., 2017, 2020; Mosayebi Samani et al., 2019; Agboada et al., 2020), and further investigation in the frontal cortex will be required to better understand these relationships and

their effects on behavior and cognitive processes (Ehrhardt et al., 2022). While the neurophysiological basis of stimulation effects is poorly understood (Filmer et al., 2020b; Bradley et al., 2022), there is growing evidence that stimulation may modulate concentrations of the neurochemicals GABA and glutamate (Stagg et al., 2009; Kim et al., 2014; Bachtiar et al., 2018; Patel et al., 2019; Filmer et al., 2023). In addition, baseline concentrations of GABA and glutamate have been associated with the efficacy of single-session prefrontal stimulation to modulate decision-making learning (Filmer et al., 2019a, 2020a). To date, the mechanisms of multisession application of tDCS are largely unknown. Here, we demonstrate that the efficacy of multisession tDCS—which can lead to transferable performance benefits—is related to baseline glutamate concentrations. Specifically, individuals with higher concentrations of glutamate showed greater transfer with right hemisphere stimulation. The neurochemical concentrations were measured in the left hemisphere, and yet it was stimulation of the right PFC that showed this relationship. This could be due to correlations between neurochemical concentrations in the two hemispheres and/or imply network-level effects between the two hemispheres. It is also possible that the return (cathode) electrode, which was placed over the left orbitofrontal cortex in the right hemisphere group, played some role in stimulation effects. However, the location of the cathode was more anterior in the frontal cortex than the MRS region of interest, so again any relationship with glutamate here would be more complex than the specific target region. In any event, our findings link the efficacy of stimulation to drive transfer effects to PFC neurochemical concentrations and provide a potential target for individually tailoring future stimulation protocols.

There is also mounting evidence that transfer can exist if the right circumstances are fulfilled (Garner et al., 2016; Smithers et al., 2018; Smid et al., 2020; Bigand and Tillmann, 2022; Garner and Dux, 2023). Here, we provide further evidence for a means of achieving transfer through specific (1 mA) stimulation to prefrontal cortices in combination with multitasking training, as well as provide insight into the neurophysiological underpinnings of the mechanisms of combined training and stimulation. Little is known about how stimulation leads to transferable performance benefits, yet this potential to generalize learning and enhance performance across multiple task types and cognitive domains holds substantial promise for a wide range of applications (Bostrom and Sandberg, 2009). It is important to note that others have found evidence of transfer effects after combining working memory training with tDCS (Pergher et al., 2022). Here, we find that excitatory processes in key higher brain regions are linked to transferable gains induced by stimulation, highlighting a key strength of MRS: it goes beyond isolating specific brain regions but rather links specific neural processes (excitatory vs inhibitory) to behavior. Increasing our understanding of the mechanisms underlying paired training and stimulation transfer and indeed stimulation generally will help enable the implementation of protocols that are optimal for enhancing performance. More broadly, investigating these processes will also contribute to our understanding of how, why, and when training leads to transferable benefits.

Data Availability

All data can be made available by contacting the corresponding author. Task scripts are available https://github.com/duxlab/tDCS_training/tree/master, and data are presented in the main text.

References

Agboada D, Mosayebi-Samani M, Kuo M-FF, Nitsche MA (2020) Induction of long-term potentiation-like plasticity in the primary motor cortex with repeated anodal transcranial direct current stimulation – better effects with intensified protocols? Brain Stimul 13:987–997.

Anderson JS, Carandini M, Ferster D (2000) Orientation tuning of input conductance, excitation, and inhibition in cat primary visual cortex. J Neurophysiol 84:909–926.

Andreychenko A, Boer VO, Arteaga de Castro CS, Luijten PR, Klomp DWJ (2012) Efficient spectral editing at 7T: GABA detection with MEGA-sLASER. Magn Reson Med 68:1018–1025.

Anguera JA, et al. (2013) Video game training enhances cognitive control in older adults. Nature 501:97–101.

Ashburner J, Friston KJ (2005) Unified segmentation. NeuroImage 26:839–851.

Bachtiar V, Johnstone A, Berrington A, Lemke C, Johansen-Berg H, Emir U, Stagg CJ (2018) Modulating regional motor cortical excitability with non-invasive brain stimulation results in neurochemical changes in bilateral motor cortices. J Neurosci 38:7327–7336.

Batsikadze G, Moliadze V, Paulus W, Kuo M-F, Nitsche MA (2013) Partially non-linear stimulation intensity-dependent effects of direct current stimulation on motor cortex excitability in humans. J Physiol 591:1987–2000.

Bender AD, Filmer HL, Naughtin CK, Dux PE (2017) Dynamic, continuous multitasking training leads to task-specific improvements but does not transfer across action selection tasks. NPJ Sci Learn 2:14.

Bigand E, Tillmann B (2022) Near and far transfer: is music special? Mem Cogn 50:339–347.

Bikson M, et al. (2016) Safety of transcranial direct current stimulation: evidence based update 2016. Brain Stimul 9:641–661.

Bikson M, Paulus W, Esmaeilpour Z, Kronberg G, Nitsche MA (2019) Mechanisms of acute and after effects of transcranial direct current stimulation. In: *Practical guide to transcranial direct current stimulation* (Knotkova H, Nitsche MA, Bikson M, Woods AJ, eds), pp 81–113. Cham: Springer International Publishing.

Bostrom N, Sandberg A (2009) Cognitive enhancement: methods, ethics, regulatory challenges. Sci Eng Ethics 15:311–341.

Bradley C, Nydam AS, Dux PE, Mattingley JB (2022) State-dependent effects of neural stimulation on brain function and cognition. Nat Rev Neurosci 23:459–475.

Brainard DH (1997) The psychophysics toolbox. Spat Vis 10:433–436.

Chua EF, Ahmed R, Garcia SM (2017) Effects of HD-tDCS on memory and metamemory for general knowledge questions that vary by difficulty. Brain Stimul 10:231–241.

Cudalbu C, et al. (2021) Contribution of macromolecules to brain 1H MR spectra: experts' consensus recommendations. NMR Biomed 34:e4393.

Dedoncker J, Brunoni AR, Baeken C, Vanderhasselt M-A (2016) A systematic review and meta-analysis of the effects of transcranial direct current stimulation (tDCS) over the dorsolateral prefrontal cortex in healthy and neuropsychiatric samples: influence of stimulation parameters. Brain Stimul 9:501–517.

Dougherty MR, Engle RW (2019) Epilogue: don't buy the snake oil. In: *Cognitive and working memory training* (Novick JM, others, eds), pp 539–544. New York: Oxford University Press.

Duncan J (2010) The multiple-demand (MD) system of the primate brain: mental programs for intelligent behaviour. Trends Cogn Sci 14:172–179.

Dux PE, Marois R (2009) The attentional blink: a review of data and theory. Atten Percept Psychophys 71:1683–1700.

Dux PE, Ivanoff J, Asplund CL, Marois R (2006) Isolation of a central bottleneck of information processing with time-resolved fMRI. Neuron 52:1109–1120.

Dux PE, Tombu MN, Harrison S, Rogers BP, Tong F, Marois R (2009) Training improves multitasking performance by increasing the speed of information processing in human prefrontal cortex. Neuron 63:127–138.

Ehrhardt SE, Filmer HL, Wards Y, Mattingley JB, Dux PE (2021) The influence of tDCS intensity on decision-making training and transfer outcomes. J Neurophysiol 125:385–397.

Ehrhardt SE, Ballard T, Wards Y, Mattingley JB, Dux PE, Filmer HL (2022) tDCS augments decision-making efficiency in an intensity dependent manner: a training study. Neuropsychologia 176:108397.

Ernst T, Kreis R, Ross BD (1993) Absolute quantitation of water and metabolites in the human brain. I. Compartments and water. J Magn Reson B 102:1–8.

Filmer HL, Mattingley JB, Marois R, Dux PE (2013) Disrupting prefrontal cortex prevents performance gains from sensory-motor training. J Neurosci 33:18654–18660.

Filmer HL, Dux PE, Mattingley JB (2014) Applications of transcranial direct current stimulation for understanding brain function. Trends Neurosci 37:742–753.

Filmer HL, Lyons M, Mattingley JB, Dux PE (2017a) Anodal tDCS applied during multitasking training leads to transferable performance gains. Sci Rep 7:12988.

Filmer HL, Varghese E, Hawkins GE, Mattingley JB, Dux PE (2017b) Improvements in attention and decision-making following combined behavioral training and brain stimulation. Cereb Cortex 27:3675–3682.

Filmer HL, Ehrhardt SE, Bollmann S, Mattingley JB, Dux PE (2019a) Accounting for individual differences in the response to tDCS with baseline levels of neurochemical excitability. Cortex 115:324–334.

Filmer HL, Griffin A, Dux PE (2019b) For a minute there, I lost myself … dosage dependent increases in mind wandering via prefrontal tDCS. Neuropsychologia 129:379–384.

Filmer HL, Ballard T, Ehrhardt SE, Bollmann S, Shaw TB, Mattingley JB, Dux PE (2020a) Dissociable effects of tDCS polarity on latent decision processes are associated with individual differences in neurochemical concentrations and cortical morphology. Neuropsychologia 141:107433.

Filmer HL, Mattingley JB, Dux PE (2020b) Modulating brain activity and behaviour with tDCS: rumours of its death have been greatly exaggerated. Cortex 123:141–151.

Filmer HL, et al. (2023) Individual differences in decision strategy relate to neurochemical excitability and cortical thickness. J Neurosci 43:7006–7015.

Framorando D, Cai T, Wang Y, Pegna AJ (2021) Effects of transcranial direct current stimulation on effort during a working-memory task. Sci Rep 11: 16399.

Garner KG, Dux PE (2015) Training conquers multitasking costs by dividing task representations in the frontoparietal-subcortical system. Proc Natl Acad Sci U S A 112:14372–14377.

Garner KG, Dux PE (2023) Knowledge generalization and the costs of multitasking. Nat Rev Neurosci 24:98–112.

Garner KG, Lynch CR, Dux PE (2016) Transfer of training benefits requires rules we cannot see (or hear). J Exp Psychol Hum Percept Perform 42: 1148–1157.

Gasparovic C, Song T, Devier D, Bockholt HJ, Caprihan A, Mullins PG, Posse S, Jung RE, Morrison LA (2006) Use of tissue water as a concentration reference for proton spectroscopic imaging. Magn Reson Med 55:1219–1226.

Gelman A, Tuerlinckx F (2000) Type S error rates for classical and Bayesian single and multiple comparison procedures. Comput Stat 15:373–390.

Heimrath K, Brechmann A, Blobel-Lüer R, Stadler J, Budinger E, Zaehle T (2020) Transcranial direct current stimulation (tDCS) over the auditory cortex modulates GABA and glutamate: a 7T MR-spectroscopy study. Sci Rep 10:20111.

Horne KS, Filmer HL, Nott ZE, Hawi Z, Pugsley K, Mattingley JB, Dux PE (2021) Evidence against benefits from cognitive training and transcranial direct current stimulation in healthy older adults. Nat Hum Behav 5:146–158.

Huang Y, Datta A, Bikson M, Parra LC (2019) Realistic volumetric-approach to simulate transcranial electric stimulation—ROAST—a fully automated open-source pipeline. J Neural Eng 16:056006.

Huang Y, Parra LC, Haufe S (2016) The New York head—a precise standardized volume conductor model for EEG source localization and tES targeting. NeuroImage 140:150–162.

Jackson MP, Rahman A, Lafon B, Kronberg G, Ling D, Parra LC, Bikson M (2016) Animal models of transcranial direct current stimulation: methods and mechanisms. Clin Neurophysiol 127:3425–3454.

Jaeggi SM, Buschkuehl M, Jonides J, Perrig WJ (2008) Improving fluid intelligence with training on working memory. Proc Natl Acad Sci U S A 105: 6829–6833.

Jaeggi SM, Buschkuehl M, Jonides J, Shah P (2011) Short- and long-term benefits of cognitive training. Proc Natl Acad Sci U S A 108:10081–10086.

Jamil A, Batsikadze G, Kuo H-II, Labruna L, Hasan A, Paulus W, Nitsche MA (2017) Systematic evaluation of the impact of stimulation intensity on neuroplastic after-effects induced by transcranial direct current stimulation. J Physiol Paris 595:1273–1288.

Jamil A, Batsikadze G, Kuo HI, Meesen RLJ, Dechent P, Paulus W, Nitsche MA (2020) Current intensity- and polarity-specific online and aftereffects of transcranial direct current stimulation: an fMRI study. Hum Brain Mapp 41:1644–1666.

Jasper H (1958) The ten-twenty electrode system of the international federation. Electroencephalogr Clin Neurophysiol 10:371–375.

Jia K, Frangou P, Karlaftis VM, Ziminski JJ, Giorgio J, Rideaux R, Zamboni E, Hodgson V, Emir U, Kourtzi Z (2022) Neurochemical and functional interactions for improved perceptual decisions through training. J Neurophysiol 127:900–912.

Jones KT, Berryhill ME (2012) Parietal contributions to visual working memory depend on task difficulty. Front Psychiatry 3:81.

Katz B, Au J, Buschkuehl M, Abagis T, Zabel C, Jaeggi SM, Jonides J (2017a) Individual differences and long-term consequences of tDCS-augmented cognitive training. J Cogn Neurosci 29:1498–1508.

Katz B, Au J, Buschkuehl M, Abagis T, Zabel C, Jaeggi SM, Jonides J (2017b) Individual differences in tDCS augmented working memory training. Brain Stimul 10:e24.

Kim S, Stephenson MC, Morris PG, Jackson SR (2014) tDCS-induced alterations in GABA concentration within primary motor cortex predict motor learning and motor memory: a 7T magnetic resonance spectroscopy study. NeuroImage 99:237–243.

Klingberg T (2010) Training and plasticity of working memory. Trends Cogn Sci 14:317–324.

Krause B, Cohen Kadosh R (2014) Not all brains are created equal: the relevance of individual differences in responsiveness to transcranial electrical stimulation. Front Syst Neurosci 8:25.

Laakso I, Mikkonen M, Koyama S, Hirata A, Tanaka S (2019) Can electric fields explain inter-individual variability in transcranial direct current stimulation of the motor cortex? Sci Rep 9:1–10.

Looi CY, Duta M, Brem A-K, Huber S, Nuerk H-C, Cohen Kadosh R (2016) Combining brain stimulation and video game to promote long-term transfer of learning and cognitive enhancement. Sci Rep 6:22003.

López-Alonso V, Cheeran B, Río-Rodríguez D, Fernández-Del-Olmo M (2014) Inter-individual variability in response to non-invasive brain stimulation paradigms. Brain Stimul 7:372–380.

Marín O (2012) Interneuron dysfunction in psychiatric disorders. Nat Rev Neurosci 13:107–120.

Marjańska M, Auerbach EJ, Valabrègue R, Van De Moortele P-F, Adriany G, Garwood M (2012) Localized ^{1}H NMR spectroscopy in different regions of human brain in vivo at 7T: t_2 relaxation times and concentrations of cerebral metabolites: regional differences in T_2 and metabolite concentrations at 7T. NMR Biomed 25:332–339.

Marois R, Ivanoff J (2005) Capacity limits of information processing in the brain. Trends Cogn Sci 9:296–305.

Melby-Lervåg M, Redick TS, Hulme C (2016) Working memory training does not improve performance on measures of intelligence or other measures of "far transfer": evidence from a meta-analytic review. Perspect Psychol Sci 11:512–534.

Miniussi C, Harris JA, Ruzzoli M (2013) Modelling non-invasive brain stimulation in cognitive neuroscience. Neurosci Biobehav Rev 37:1702–1712.

Mosayebi Samani M, Agboada D, Jamil A, Kuo M-F, Nitsche MA (2019) Titrating the neuroplastic effects of cathodal transcranial direct current stimulation (tDCS) over the primary motor cortex. Cortex 119:350–361.

Nitsche MA, Paulus W (2000) Excitability changes induced in the human motor cortex by weak transcranial direct current stimulation. J Physiol 527:633–639.

Nitsche MA, Paulus W (2001) Sustained excitability elevations induced by transcranial DC motor cortex stimulation in humans. Neurology 57: 1899–1901.

Owen AM, Hampshire A, Grahn JA, Stenton R, Dajani S, Burns AS, Howard RJ, Ballard CG (2010) Putting brain training to the test. Nature 465:775–778.

Patel HJ, Romanzetti S, Pellicano A, Nitsche MA, Reetz K, Binkofski F (2019) Proton magnetic resonance spectroscopy of the motor cortex reveals long term GABA change following anodal transcranial direct current stimulation. Sci Rep 9:2807.

Pelli DG (1997) The VideoToolbox software for visual psychophysics: transforming numbers into movies. Spat Vis 10:437–442.

Pergher V, Au J, Alizadeh Shalchy M, Santarnecchi E, Seitz A, Jaeggi SM, Battelli L (2022) The benefits of simultaneous tDCS and working memory training on transfer outcomes: a systematic review and meta-analysis. Brain Stimul 15:1541–1551.

Petroff OAC (2002) GABA and glutamate in the human brain. Neuroscientist 8:562–573.

Polanía R, Nitsche MA, Ruff CC (2018) Studying and modifying brain function with non-invasive brain stimulation. Nat Neurosci 21:174–187.

Poo C, Isaacson JS (2009) Odor representations in olfactory cortex: "sparse" coding, global inhibition, and oscillations. Neuron 62:850–861.

Priori A, Berardelli A, Rona S, Accornero N, Manfredi M (1998) Polarization of the human motor cortex through the scalp. Neuroreport 9:2257–2260.
Provencher SW (1993) Estimation of metabolite concentrations from localized in vivo proton NMR spectra. Magn Reson Med 30:672–679.
Rango M, Cogiamanian F, Marceglia S, Barberis B, Arighi A, Biondetti P, Priori A (2008) Myoinositol content in the human brain is modified by transcranial direct current stimulation in a matter of minutes: a 1 H-MRS study. Magn Reson Med 60:782–789.
Redick TS, Shipstead Z, Harrison TL, Hicks KL, Fried DE, Hambrick DZ, Kane MJ, Engle RW (2013) No evidence of intelligence improvement after working memory training: a randomized, placebo-controlled study. J Exp Psychol Gen 142:359–379.
Reinhart RMG, Cosman JD, Fukuda K, Woodman GF (2017) Using transcranial direct-current stimulation (tDCS) to understand cognitive processing. Atten Percept Psychophys 79:3–23.
Ridding MC, Ziemann U (2010) Determinants of the induction of cortical plasticity by non-invasive brain stimulation in healthy subjects. J Physiol 588:2291–2304.
Rideaux R (2020) Temporal dynamics of GABA and Glx in the visual cortex. eNeuro 7:1–11.
Rideaux R (2021) No balance between glutamate+glutamine and GABA+ in visual or motor cortices of the human brain: a magnetic resonance spectroscopy study. NeuroImage 237:118191.
Rideaux R, Goncalves NR, Welchman AE (2019) Mixed-polarity random-dot stereograms alter GABA and Glx concentration in the early visual cortex. J Neurophysiol 122:888–896.
Rideaux R, Ehrhardt SE, Wards Y, Filmer HL, Jin J, Deelchand DK, Marjańska M, Mattingley JB, Dux PE (2022) On the relationship between GABA+ and glutamate across the brain. NeuroImage 257:119273.
Rideaux R, Mikkelsen M, Edden RAE (2021) Comparison of methods for spectral alignment and signal modelling of GABA-edited MR spectroscopy data. NeuroImage 232:117900.
Sala G, Gobet F (2017) Does far transfer exist? Negative evidence from chess, music, and working memory training. Curr Dir Psychol Sci 26:515–520.
Sala G, Gobet F (2019) Cognitive training does not enhance general cognition. Trends Cogn Sci 23:9–20.
Scheenen TWJ, Heerschap A, Klomp DWJ (2008) Towards 1H-MRSI of the human brain at 7T with slice-selective adiabatic refocusing pulses. Magn Reson Mater Phys Biol Med 21:95–101.
Shadlen MN, Newsome WT (1994) Noise, neural codes and cortical organization. Curr Opin Neurobiol 4:569–579.
Shipstead Z, Redick TS, Engle RW (2012) Is working memory training effective? Psychol Bull 138:628–654.
Simons DJ, Boot WR, Charness N, Gathercole SE, Chabris CF, Hambrick DZ, Stine-Morrow EAL (2016) Do "brain-training" programs work? Psychol Sci Public Interest 17:103–186.
Smid CR, Karbach J, Steinbeis N (2020) Toward a science of effective cognitive training. Curr Dir Psychol Sci 29:531–537.
Smithers LG, Sawyer ACP, Chittleborough CR, Davies NM, Davey Smith G, Lynch JW (2018) A systematic review and meta-analysis of effects of early life non-cognitive skills on academic, psychosocial, cognitive and health outcomes. Nat Hum Behav 2:867–880.
Stagg CJ (2014) The physiological basis of brain stimulation.
Stagg CJ, Best JG, Stephenson MC, O'Shea J, Wylezinska M, Kincses ZT, Morris PG, Matthews PM, Johansen-Berg H (2009) Polarity-sensitive modulation of cortical neurotransmitters by transcranial stimulation. J Neurosci 29:5202–5206.
Tkáć I, Gruetter R (2005) Methodology of 1H NMR spectroscopy of the human brain at very high magnetic fields. Appl Magn Reson 29: 139–157.
Trumbo MC, Matzen LE, Coffman BA, Hunter MA, Jones AP, Robinson CSH, Clark VP (2016) Enhanced working memory performance via transcranial direct current stimulation: the possibility of near and far transfer. Neuropsychologia 93:85–96.
Unsworth N, Heitz RP, Schrock JC, Engle RW (2005) An automated version of the operation span task. Behav Res Methods 37:498–505.
van Doorn J, et al. (2020) The JASP guidelines for conducting and reporting a Bayesian analysis. Psychon Bull Rev 28:813–826.
van Vreeswijk C, Sompolinsky H (1996) Chaos in neuronal networks with balanced excitatory and inhibitory activity. Science 274:1724–1726.
Vergallito A, Feroldi S, Pisoni A, Romero Lauro LJ (2022) Inter-individual variability in tDCS effects: a narrative review on the contribution of stable, variable, and contextual factors. Behav Brain Sci 12:522.
Vogels TP, Abbott LF (2009) Gating multiple signals through detailed balance of excitation and inhibition in spiking networks. Nat Neurosci 12:483–491.
Wagenmakers E-J, et al. (2018) Bayesian inference for psychology. Part II: example applications with JASP. Psychon Bull Rev 25:58–76.
Wards Y, Ehrhardt SE, Filmer HL, Mattingley JB, Garner KG, Dux PE (2023) Neural substrates of individual differences in learning generalization via combined brain stimulation and multitasking training. Cereb Cortex 33:11679–11694.
Wehr M, Zador AM (2003) Balanced inhibition underlies tuning and sharpens spike timing in auditory cortex. Nature 426:442–446.
Whelan R (2008) Effective analysis of reaction time data. Psychol Rec 58:475–482.
Wilent WB, Contreras D (2005) Dynamics of excitation and inhibition underlying stimulus selectivity in rat somatosensory cortex. Nat Neurosci 8:1364–1370.
Wolfe JM, Horowitz TS (2004) What attributes guide the deployment of visual attention and how do they do it? Nat Rev Neurosci 5:495–501.
Wörsching J, Padberg F, Ertl-Wagner B, Kumpf U, Kirsch B, Keeser D (2016) Imaging transcranial direct current stimulation (tDCS) of the prefrontal cortex—correlation or causality in stimulation-mediated effects? Neurosci Biobehav Rev 69:333–356.
Xin L, Schaller B, Mlynarik V, Lu H, Gruetter R (2013) Proton T_1 relaxation times of metabolites in human occipital white and gray matter at 7T: metabolites T_1 relaxation times in human brain at 7T. Magn Reson Med 69:931–936.